Coefficient Extraction Operators in the Theory of Multidimensional Random Walk Enumeration

by

Aleksandar Vlasev

B.Sc. Mathematics, Simon Fraser University, 2012

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

Master of Science

in

THE FACULTY OF SCIENCE
(Mathematics)

The University of British Columbia
(Vancouver)

April 2015

© Aleksandar Vlasev, 2015

Abstract

In this work we explore the the application of operators to enumeration of multidimensional random walks with a focus on walks which interact with the boundary hyperplanes. Using operators makes the derivation of the functional equations a lot easier and some preliminary work shows that using operators is the way to go when solving problems of this nature.

Up to 350 words

Preface

This work intends to establish a new framework of notation and tools for the enumeration of random walks with or without interactions, and similar problems. The notation makes it much easier to explore the symmetries of the problem and write down summation formulas where it was not possible to do so before (with a simplified notation). The tools make the derivation of functional equations for various problems an almost trivial matter. The real power is in combining both approaches to gain insights into the generating function world of enumerating random walks on constrained regions in multidimensional spaces.

Table of Contents

Abstract . . . . . **ii**

Preface . . . . . **iii**

Table of Contents . . . . . **iv**

List of Tables . . . . . **viii**

List of Figures . . . . . **ix**

Glossary . . . . . **x**

Acknowledgments . . . . . **xi**

1 Introduction **1**

2 One-dimensional random walks **5**

2.1 Derivation of the functional equation 5

2.2 Solving the functional equation for various s 11

2.2.1 The $s_{-1} = 0$ case. 12

2.2.2 The $s_{-1} \neq 0$ case. 13

2.2.3 1-dimensional case conclusion 17

3 Two-dimensional Random Walks **18**

3.1 Introduction 18

3.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$ 20

3.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning . . . 21
3.4 Simple case: Factoring S into a product . . . 22
3.5 Kernel Group Considerations . . . 25
3.6 Solving basic 2D models . . . 27
3.6.1 Cases 1.1, 1.2, 1.3 . . . 27
3.6.2 Case 2.2 - group of order 1 . . . 28
3.6.3 Case 2.3 - group of order 2 . . . 31

4 Hexagonal Walks . . . 33
4.1 Introduction . . . 33
4.2 Operators . . . 35
4.3 Group Structure . . . 36
4.4 Finding the generating function Q . . . 40

5 Higher-dimensional walks . . . 42
5.1 Using Operators to Derive the Functional Equation . . . 42
5.2 Coefficient Extraction . . . 44
5.3 Further functional equations . . . 46
5.4 Using the formal inverse for iteration . . . 48
5.5 Kernel Form . . . 49
5.6 Walks without interactions . . . 50

6 Operators . . . 53
6.1 Homogeneous Operators . . . 53
6.1.1 Single 1D Operators . . . 53
6.1.2 Compounded 1D Operators . . . 55
6.1.3 Multidimensional Operators . . . 56
6.2 Inhomogeneous Operators . . . 56
6.2.1 Single 1D Operators . . . 57
6.3 Random Walks Operator . . . 59
6.4 Interaction with ∂_t . . . 60
6.5 Interaction with ∂_i . . . 60

7 Discussion on Random Walk Model Generalizations 62
7.1 Introduction 62
7.2 Marking Operators 65
7.3 Self-avoiding walks 68
7.3.1 Eigenpairs 71
7.3.2 Generalizations 72
7.4 Coefficient Extraction Equations 72
7.5 Connection with Fredholm Theory 77

8 Groups for walks with small steps 80
8.1 Coxeter Groups 81
8.2 Orbit Sums and Kernel Method 83
8.3 Half-Orbit Sums 86
8.4 Infinite Group Orbits 86

9 Orbit Summation for Walks with Interactions 87
9.1 Group of order 4 87
9.1.1 When is $\det M_4 = 0$? 89
9.1.2 Analogue of the Orbit Summation Method 91
9.1.3 Similarities between labeled and unlabeled versions 94
9.2 Group of order 6 95
9.3 Group of order 8 97
9.4 Infinite Groups 98

10 Highly-symmetric 2D models 99
10.1 Cartesian model: steps N, S, E, W 101
10.2 Diagonal model: steps NE, NW, SE, SW 104
10.3 The almost-full model: missing steps E and W 113
10.4 The full model: all steps available 113
10.5 Conjectured relationships depending on $S_{1,2}^{-1,-1}$ 113

11 Epilogue 114
11.1 Discussion 114
11.2 Future Work 114

11.3 Conclusion 114

12 Miscellaneous Things 115

A Supporting Materials 118

List of Tables

List of Figures

Glossary

This glossary uses the handy `acroynym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the LaTeX source.

Acknowledgments

I wish to say thank you to everyone who has helped me along my graduate school journey.

Thank you Victoria Zhu for supporting me throughout all these years. Without you love and support I would have quit many times! You were always there for me when I had doubts and wanted to give up.

Thank you Andrew Rechnitzer for taking me under your wing and believing in me. You are the man!

Thank you Kalle Karu. Without your support I wouldn't have made it this far. Your generousity will be forever appreciated!

Thank you mom and dad for believing in me, guiding me and supporting me in all the times of need. There's nothing quite like hearing you say that I can do it!

Thank you Tony Chong, Gloria Chien and Vlad Bohdanov for being my coaches and mentors, and helping me on the path to become a champion!

Also a big thank you to all my friends for letting me rant so many times!

Last but not least I want to thank UBC and the mathematics department especially for supporting me through thick and thin and somehow making things work for me whenever all hope was lost.

If I forgot anybody...you can just kill me.

Chapter 1

Introduction

In this work we'll attempt to further the mathematics of the enumeration of discrete random walks. Typically those problems take place on a cubical lattice and one may often investigate random walk models with different steps and various constraints to smaller regions. In this work we'll tackle the problem of random walk models with interactions and specifically counting these interactions. But before we get to it we'll go through some of the history of the problem.

As with any enumeration problem we start simple with a two-dimensional random walk model with possible steps - up, down, left and right. A typical question we can answer is how many walks of length n are there? In this case we have 4^n. We can compe up with more complicated models with various step lengths and directions. There has been a lot of work done in restricting such models to parts of the plane - half-plane and quarter plane. There have been models on semi-infinite strips and wedges. There have been generalizations to more dimensions like random walks in the positive orthant, sometimes in the full generality of d dimensions. Another type of discrete random walk model is that of self-avoiding walks. One may investigate these on the plane or on a different lattice (like a honeycomb lattice). Further we may try to answer questions about walks interacting with interfaces - quarterplane walks interacting with the axes or a pair of random walks interacting with each other. In this work we'll mainly investigate random walks with interactions in 2 or more dimensions. In a few cases we'll touch upon random walks on other lattices and self-avoiding walks.

The natural question is why study these objects in the first place? There are natural applications in the world of biology where one can investigate simple protein models - chains of atoms. Self-avoiding walks naturally fit into this narrative. If one wishes to model interactions between proteins (or other chains) with some interface, one can use interacting random walks in order to gain insight into the large-scale properties of protein-like structure. Mathematically speaking, the subject of random walk enumeration is rich. The main tool is the generating function tracking various parameters of random walks - length, coordinates, number of interactions with other objects, etc. In two or more dimensions we find a rich theory of functional equations involving these generating functions. The solutions to those equations often depend sensitively to the step set present and on the geometry of the region involved. Interactions tend to complicate the matter even further.

One of the most pertinent questions in this field has to do with the D-finiteness of the random walk generating functions in the variable keeping track of length (typically t). If a generating function is D-finite we can obtain a differential equation for it and perhaps solve it explicitly. It turns out that many interesting models lead to generating functions that are not D-finite. There has been a full classification of two-dimensional random walks without interactions on the full plane, half plane and quarter plane. There has been a partial classification for three-dimensional random walks without interactions for step sets of size at most 6. Such classification has not been performed for random walks with interactions. Although it's worth mentioning that when a model with no interaction leads to a non-D-finite generating function, then the generating function for the interacting case must also be non-D-finite. It's probably a good guess that there are D-finite non-interacting models which has a non-D-finite interacting counterpart. Typical tools for obtaining information about the random walk generating function are group theory for obtaining more functional equations or a simplified one and complex analysis for solving problems pertaining to D-finiteness. More specifically we have the orbit method, the vanishing kernel method, iteration of roots and so on.

So far most discrete random walk models have not needed much special notation or special tools in order to derive functional equations for the generating functions involved and solve them. This has been true for the non-interacting two and three-dimensional problems on the full, half, quarter, etc regions. However,

if we are to tackle more general problems like walks with interactions in more dimensions and/or domains of interest we need more powerful notation. Even deriving the two-dimensional functional equations for walks with interactions is pain-staking. Putting it into a simpler form can be a time-consuming task.

A dreamy goal is to make the process fully automatic. We should be able to input a random walk model (step set, domain, interactions) into an algorithm that will output several items in order of increasing utility - a functional equation for the generating function, a reduced such equation for easier solving and hopefully a solution for the equation and/or showing that it is/isn't D-finite. In this work we achieve part of the goal - the first two outputs of this algorithm. Some of the main results are that for a lot of random walk problems we can write an operator equation for the generating function that we can either solve iteratively or simplify to a form more conducive to further analysis with group theory and other tools. The work hints at a deeper theory of coefficient extraction equations which are similar to differential equations. Instead of derivatives we have coefficient extractions. The connection is made exact by noting that coefficient extraction is the same as taking a derivative and evaluating at zero for that specific variable, modulo a constant. Surely we should see further development in a theory of ordinary and partial coefficient extraction equations.

In the first few chapters we go through motivating examples which will serve as a springboard into further generalizations. First we'll start off with one-dimensional random walks with and without interactions in Chapter 2. Then we'll quickly move on to two-dimensional random walks in Chapter 3. On the way we'll develop some of the basic operator theory involved in the random walk models. In Chapter 4 we'll make a detour to random walks on a honeycomb lattice - a problem that will show some of the group-theoretic and operator tools. In Chapter 5 we'll tackle the d-dimensional random walk model functional equations using operators. In chapter 6 we'll finally go through operators in a much generality as we can, deriving all sorts of properties for them along the way. In chapter 7 we'll discuss generalizations of the random walk models and we'll push the operator notation as far as we can to see what kind of problem we can hope to tackle with the theory.

In Chapter 8 we'll become more practical and deal with group-theoretic operators relevant to both two-dimensional and higher-dimensional random walk mod-

els. In Chapter 9 we extend the persistent kernel method as much as we can for the two-dimensional interacting walks. Then in Chapter 10 we attempt to solve as many highly-symmetrical two-dimensional interacting models as we can via the vanishing kernel method. In Chapter 11 we work on some models with higher group orders.

Chapter 2

One-dimensional random walks

2.1 Derivation of the functional equation

In the interest of working out higher dimensional walk problems it is very useful to first visit the 1-dimensional case which will illustrate a lot of the methods that will later come into play. The 1-dimensional case will also give us a clearer picture as to how things work in higher dimensions (although the picture becomes more complicated). We'll treat the problem using generating functions. We'll use lower case letters and not quite the full notation because it is not necessary. In one dimension, on one axis, we'll use steps of length at most 1. If we have a random walk ending at coordinate i we can either step back adding -1 to the coordinate, step forward adding $+1$ to the coordinate or staying in one place, adding 0 to the coordinate. This can be somewhat formalized by using polynomials. Ending a walk at coordinate i will be encoded with a monomial x^i. Then stepping backwards, forwards or staying in one place will be denoted with x^{i-1}, x^i and x^{i+1} respectively. This is the same as multiplying by $1/x = \overline{x}, 1, x$ in each case. We can encode this information is a Laurent step polynomial s defined as

$$s = s_{-1}\overline{x} + s_0 + s_1 x \tag{2.1}$$

Here s_i will be either 0 or 1 for any specific i, tracking whether we have allowed those kinds of steps or not. Keeping this in mind, there are 8 possible step sets

because we have two options for each s_i. We keep track of the length of a random walk with powers of t and we keep track of its endpoint with powers of x. We'll also mark every interaction with the origin by multiplying by a every time a walk steps on the origin. We'll keep the walks on the non-negative axis.

Long story short we can find the generating function in a very simple, brute force manner, even for higher dimensions. We start with the empty walk denoted by 1 since it starts and ends on the origin without taking a step, so no t, x or mark a. Then try out each step from s and eliminate those that take us past the origin in the negative direction. If we step on the origin we multiply by a. Those walks that remain are then multiplied by t since they are not of length 1. We can proceed iteratively k tiems to obtain random walks on the non-negative axis of length k. Adding up the results allows us to write the generating function up to $O(t^k)$. It will prove very useful to think about operators so we can formalize this notion in terms of operators. For example, here we are taking walks of length $k-1$, trying all options of s (this is like multiplying by s), removing all the walks that end up on the negative axis (this is like removing the $\bar{x}$ term) and marking those that end on the origin (this is like multiplying the x^0 coefficient by a). Let D stand for the operator that cleans up the walks and marks the valid ones. Then if q_k stands for the marked walks of length k, then the k-th term of q is $q_k t^k$ and we have

$$q_{k+1}t^{k+1} = tD(sq_k t^k) \tag{2.2}$$

Thus we can write the generating function q iteratively

$$q = \sum_{k=0}^{\infty} t^k (Ds)^k (1) \tag{2.3}$$

where we're iteratively applying Ds to 1 to obtain all marked walks. We can treat this operator Ds as a formal variable and write

$$q = \sum_{k=0}^{\infty} (tDs)^k (1) = \frac{1}{1-tDs}(1) \tag{2.4}$$

and then this gives us a functional equation for q

$$(1 - tDs)q = 1 \tag{2.5}$$

There is another way of deriving this equation while not treating Ds as a formal variable. Instead of applying the operator iteratively we can apply it on the generating function as a whole. The product tsq is the generating for all random walks that end with x-coordinate greater or equal to -1 where some steps end up on x-coordinate 0 without being marked. We can apply D on this product to remove the terms ending at x-coordinate -1 and mark those that end up on x-coordinate 0. This leaves us with all the valid walks encoded in the generating function q without the empty walk 1. Therefore we can write a functional equation for q as

$$q = 1 + D(tsq) = 1 + tDs(q) \tag{2.6}$$

and we ca re-write this as $(1 - tDs)q = 1$ as before. Now, we have to say a few words about what the operator D actually is. First we are removing all the walks that end on x-coordinate equal to -1 which can be handled by the operator $E = 1 - \overline{x}[\overline{x}]$ which removes those walks. Then the markling can be handled by the operator $L = 1 + (a-1)[x^0] = 1 + b[x^0]$ where $b = a - 1$. Therefore we can write $D = LE$

$$D = LE = (1 - \overline{x}[\overline{x}])(1 + b[x^0]) \tag{2.7}$$

$$= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \tag{2.8}$$

$$= 1 + b[x^0] - \overline{x}[\overline{x}] \tag{2.9}$$

since the last term vanishes on all generating functions. We could have also applied the operators in reverse to get the same result.

Example 1. Suppose that $s = x + \bar{x}$. Then starting with $q_0 = 1$

$$q_1 = LEs(1) = (1 + b[x^0] - \bar{x}[\bar{x}])(x + \bar{x}) = x + \bar{x} - \bar{x} = x \tag{2.10}$$

$$q_2 = LEs(q_1) = (1 + b[x^0] - \bar{x}[\bar{x}])(x^2 + 1) = x^2 + 1 + b = a + x^2 \tag{2.11}$$

$$q_3 = LEs(q_2) = (1 + b[x^0] - \bar{x}[\bar{x}])(a\bar{x} + ax + x + x^3) \tag{2.12}$$

$$= a\bar{x} + ax + x + x^3 - a\bar{x} = (1 + a)x + x^3 \tag{2.13}$$

Therefore

$$q = q_0 + q_1 t + q_2 t^2 + q_3 t^3 + \ldots \tag{2.14}$$

$$= 1 + xt + (a + x^2)t^2 + ((1 + a)x + x^3)t^3 + \ldots \tag{2.15}$$

This commutativity makes sense because on the level of the function they are operator interacting with different monomials. We can now expand the functional equation to the following

$$1 = q - tLE(sq) = q - t(sq + b(sq)_0 - \bar{x}(sq)_{-1}) \tag{2.16}$$

$$= q - t(sq + b(s_{-1}q_1 + s_0 q_0 + s_1 q_{-1}) - \bar{x}(s_{-1}q_0)) \tag{2.17}$$

$$= q - t(sq + b(s_{-1}q_1 + s_0 q_0) - \bar{x}(s_{-1}q_0)) \tag{2.18}$$

since $q_{-1} = 0$. We can rewrite this equation in the following form

$$(1 - ts)q - t(bs_0 - \bar{x}s_{-1})q_0 - bts_{-1}q_1 = 1 \tag{2.19}$$

Here $q_0 = q(0)$ and $q_1 = q'(0)$ where $q = q(x)$. In the interest of simplifying the equation we can eliminate the q_1 term by extracting the x^0 coefficient from the

whole equation

$$1 = q_0 - t(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - tbs_0q_0 - bts_{-1}q_1 \tag{2.20}$$
$$= q_0 - t(s_{-1}q_1 + s_0q_0) - tbs_0q_0 - bts_{-1}q_1 \tag{2.21}$$
$$= (1 - ts_0 - tbs_0)q_0 - (ts_{-1} + bts_{-1})q_1 \tag{2.22}$$
$$= (1 - ats_0)q_0 - t(s_{-1} + bs_{-1})q_1 \tag{2.23}$$
$$= (1 - ats_0)q_0 - ats_{-1}q_1 \tag{2.24}$$

We can solve this equation for q_1

$$ts_{-1}q_1 = \overline{a}((1 - ats_0)q_0 - 1) \tag{2.25}$$

and substitute the result into the functional equation (for now we don't worry about the possibility of $s_{-1} = 0$ because it leads to the same thing anyway).

$$1 = (1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - c((1 - ats_0)q_0 - 1) \tag{2.26}$$
$$= (1 - ts)q - (c(1 - ats_0) + bts_0 - t\overline{x}s_{-1})q_0 + c \tag{2.27}$$
$$= (1 - ts)q - (c - bts_0 + bts_0 - t\overline{x}s_{-1})q_0 + c \tag{2.28}$$
$$= (1 - ts)q - (c - t\overline{x}s_{-1})q_0 + c \tag{2.29}$$
$$\tag{2.30}$$

and therefore we can rewrite the equation as

$$(1 - ts)q - (c - t\overline{x}s_{-1})q_0 = \overline{a} \tag{2.31}$$

since $1 - c = 1 - (1 - \overline{a}) = \overline{a}$. This process is rather complicated, even though it can be done for higher dimensional problems as well (as you can see in various previous approaches). Also there is an issue that we need to consider every time we are solving an equation in order to substitute quantities back into the equation. This is where the operator notation comes to the rescue. It turns out the the L operator is invertible with inverse given by

$$L^{-1} = M = 1 - c[x^0] \tag{2.32}$$

and we can apply it on the functional equation. The RHS is simply $M(1) = 1 - c = \overline{a}$ and the LHS is

$$M(1 - tLEs)q = Mq - tEsq \tag{2.33}$$
$$= (q - cq_0) - t(sq - \overline{x}s_{-1}q_0) \tag{2.34}$$
$$= (1 - ts)q - (c - t\overline{x}s_{-1})q_0 \tag{2.35}$$

and we've derived the same equation but in a much neater way.

Proposition 2. *The one-dimensional random walk model with laurent step polynomial $s = s_{-1}\overline{x} + s_0 + s_1 x$ has a generating function q satisfying the functional equation*

$$(M - tEs)q = (1 - ts)q - (c - t\overline{x}s_{-1})q_0 = \overline{a} \tag{2.36}$$

This will prove crucial later on when we derive a functional equation for higher dimensional problems. Although it can be done directly with a lot of effort it's much better to just use the operator notation. The derivation is brought down from about a dozen pages to one or two. The operator notation will also allow us to solve some multidimensional problems directly or with the help of lower dimensional problems. We can extract the x^0 coefficient from the final functional equation again and although we can do this directly, we'll use the operator notation to tidy up the process. This will show up later when dealing with higher dimensional problems

$$[x^0]M = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0] \tag{2.37}$$
$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0] \tag{2.38}$$

Then we extract the x^0 from the functional equation

$$[x^0]\overline{a} = \overline{a} = [x^0](M - tEs)q = ([x^0]M - t[x^0]Es)q = (\overline{a}[x^0] - t[x^0]s)q = [x^0](\overline{a} - ts)q \tag{2.39}$$

Therefore the extraction leads to

$$[x^0](1-ast)q = 1 \tag{2.40}$$

We could extract higher coefficients from the functional equation too. Let $k \geq 1$. Then

$$[x^k]M = [x^k] - [x^k]c[x^0] = [x^k] \tag{2.41}$$

$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k] \tag{2.42}$$

Therefore

$$[x^k]\overline{a} = 0 = [x^k](M - tEs)q = ([x^k]M - t[x^k]Es)q = ([x^k] - t[x^k]s)q = [x^k](1-ts)q \tag{2.43}$$

That is

$$[x^k](1-ts)q = 0 \tag{2.44}$$

and we can readily extract whichever coefficients we want. The first equation leads to

$$(1-ats_0)q_0 - ats_{-1}q_1 = 1 \tag{2.45}$$

and the second leads to

$$q_k - t(s_{-1}q_{k+1} + s_0q_k + s_1q_{k-1}) = 0 \tag{2.46}$$

$$(1-s_0t)q_k - t(s_{-1}q_{k+1} + s_1q_{k-1} = 0 \tag{2.47}$$

2.2 Solving the functional equation for various s

In this section we'll solve all 8 one-dimensional problems fully. Some cases are easier to deal with than others. As we'll see later in the higher dimensional cases, some problems are easierto solve than others. This difficulty tends to scale up with the inclusion of certain steps with negative direction.

2.2.1 The $s_{-1} = 0$ case.

This condition allows us to solve explicitly for q_0.

$$q_0 = \frac{1}{1 - as_0 t} \tag{2.48}$$

We can substitute this expression in the functional equation and obtain

Proposition 3. *When $s_{-1} = 0$ we have*

$$q(x, a; t) = \frac{\overline{a}}{1 - st}\left(1 + \frac{b}{1 - as_0 t}\right) = \frac{1 - s_0 t}{(1 - st)(1 - as_0 t)} \tag{2.49}$$

This takes care of half of the cases! What happens when we set $a = 1$, corresponding to setting $b = 0$ or extracting the b^0 coefficient from the equation? This corresponds to unmarkling the random walks. We obtain

$$[b^0]q = \widehat{q} = \frac{1 - s_0 t}{(1 - st)(1 - s_0 t)} \tag{2.50}$$

We'll note that

$$(1 - as_0 t)q - (1 - s_0 t)\widehat{q} = 0 \tag{2.51}$$

This can also be written in operator form with $B = 1 - [b^0]$

Proposition 4. *When $s_{-1} = 0$ we have*

$$B((1 - as_0 t)q) = 0 \tag{2.52}$$

This relates the marked generating function to the unmarked generating function! it says we can obtain the marked generating function from the unmarked one by the following relation

$$q = \frac{1 - as_0 t}{1 - s_0 t}\widehat{q} \tag{2.53}$$

We can also establish such as relationship for q_0 although the derivation here will

be trivial. Extracting $[x^0]$ from the equation leads to

$$B((1-as_0t)q_0) = 0 \tag{2.54}$$

and

$$q_0 = \frac{1-as_0t}{1-s_0t}\widehat{q_0} \tag{2.55}$$

The relationship between q and $\widehat{q}$ is no accident but it tends to happen fairly often in the higher dimensional cases that we've considered. The functional equation in the $s_{-1} = 0$ case is easier to solve most likely because when $s_{-1} = 0$, walks don't go beyond the origin in the negative direction. Consequently the E operator acts as the identity on the generating function. and the functional equation reads as

$$(M-ts)q = q - cq_0 - tsq = (1-ts)q - cq_0 = \overline{a} \tag{2.56}$$

2.2.2 The $s_{-1} \neq 0$ case.

The functional equation in the $s_{-1} \neq 0$ is also not difficult to solve. We'll try to apply the kernel method whenever possible. It requires finding an x value that makes the kernel $1-ts$ equal to zero in the functional equation

$$(1-ts)q - (c-t\overline{x})q_0 = \overline{a} \tag{2.57}$$

Such a solution will exist since $s_{-1} = 1$ and therefore the equation $1-ts = 0$ is non-trivial. Let one such solution be $x = \sigma$ and suppose that we can plug it into the functional equation. Then the kernel vanishes, taking q away with it and we obtain

$$-(c-t\overline{\sigma})q_0 = \overline{a} \tag{2.58}$$

We can solve for q_0 explicitly again

$$q_0 = \frac{\overline{a}}{\overline{\sigma}t - c} \tag{2.59}$$

We can now solve for q

$$q = \frac{\overline{a}}{1-ts}\left(1 - \frac{t\overline{x}-c}{t\overline{\sigma}-c}\right) = \frac{t}{1-ts}\left(\frac{\overline{x}-\overline{\sigma}}{b-a\overline{\sigma}t}\right) \tag{2.60}$$

We note that we can apply the B operator again

$$B((a\overline{\sigma}t-b)q) = 0 \tag{2.61}$$

Proposition 5. *When $s_{-1} \neq 0$ we have*

$$q = \frac{t}{1-ts}\left(\frac{\overline{x}-\overline{\sigma}}{b-a\overline{\sigma}t}\right) \tag{2.62}$$

$$0 = B((a\overline{\sigma}t-b)q) \tag{2.63}$$

Written out explicitly

$$(a\overline{\sigma}t-b)q - (\overline{\sigma}t)\widehat{q} = 0 \tag{2.64}$$

This also allows us to write the marked generating function in terms of the unmarked generating function

$$q = \frac{\overline{\sigma}t}{a\overline{\sigma}t-b}\widehat{q} = \frac{t}{at-b\sigma}\widehat{q} \tag{2.65}$$

We can also work with q_0 at this stage. We find

$$[x_0]B((a\overline{\sigma}t-b)q) = B((a\overline{\sigma}t-b)q_0) = 0 \tag{2.66}$$

too. This ends up being the same as

$$(a\overline{\sigma}t-b)q_0 - (\overline{\sigma}t)\widehat{q_0} = 0 \tag{2.67}$$

with solution

$$q_0 = \frac{t}{at-b\sigma}\widehat{q_0} \tag{2.68}$$

The final task for this section will be to find what σ will be for various cases. We

already know $s_{-1} = 0$ so the result depend on s_0 and s_1. In general the kernel is zero when

$$ts = 1 \tag{2.69}$$

$$t(s_{-1} + s_0 x + s_1 x^2) = x \tag{2.70}$$

$$ts_1 x^2 - (1 - s_0 t)x + ts_{-1} = 0 \tag{2.71}$$

When $s_1 \neq 0$ this is quadratic with two roots that are functions of t. We have to pick the one which has no singularity at $t = 0$. When $s_1 = 0$ we have a linear equation which may have a single root. When $s_1 = 0$ we have the equation $(1 - s_0 t)x = t$ so the single root is

$$\sigma = \frac{t}{1 - s_0 t} \tag{2.72}$$

When $s_1 = 1$ we have the equation $tx^2 - (1 - s_0 t)x + t = 0$ with roots

$$\sigma = \frac{1 - s_0 t - \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}, \quad \sigma^* = \frac{1 - s_0 t + \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t} \tag{2.73}$$

The two roots multiply to 1 so $\overline{\sigma} = \sigma^*$. Later we shall have a little table tabulating all the results. Note that in the relationship between the marked and unmarked q_0 we have

$$q_0 = \frac{t}{at - b\sigma} \widehat{q_0} \tag{2.74}$$

where σ is a function containing a square root, i.e. not a polynomial. However we can go at the problem from a different angle to obtain a polynomial $f(a,t)$ for which $B(fq_0) = c$ which is not quite zero but it's a simpler relationship. This will also illustrate the group approach together with the kernel method. The two cases that we need to work on both have $s_{-1} = s_1 = 1$ with s_0 being 0 or 1. This means that s is left invariant under sending $* : x \mapsto \bar{x}$. We can apply this on the functional

equation and now have a system of two functional equations

$$(1-ts)q-(c-t\overline{x})q_0-\overline{a}=0 \tag{2.75}$$

$$(1-ts)q^*-(c-tx)q_0-\overline{a}=0 \tag{2.76}$$

We can multiply the first by $c-tx$, the second by $c-t\overline{x}$ and subtract the equations to eliminate the q_0 term. We can divide everything out by $1-ts$ and finally obtain

$$(c-tx)q(x)-(c-t\overline{x})q(\overline{x})-\frac{(c-tx-c+t\overline{x})}{1-ts}\overline{a}=0 \tag{2.77}$$

$$(c-tx)q(x)-(c-t\overline{x})q(\overline{x})-\overline{a}\delta=0 \tag{2.78}$$

where

$$\delta=\frac{t(\overline{x}-x)}{1-ts} \tag{2.79}$$

and crucially it doesn't depent on a. We extract the x^1 coefficient from the equation to obtain an equation relating q_1 and q_0

$$cq_1-tq_0-\overline{a}\delta_1=0 \tag{2.80}$$

We can set $a=1$ here so we get

$$-\delta_1=t\widehat{q_0} \tag{2.81}$$

and we can bring this back to the original equation to get

$$cq_1=tq_0-\overline{a}t\widehat{q_0} \tag{2.82}$$

$$bq_1=t(aq_0-\widehat{q_0}) \tag{2.83}$$

Earlier we had the equation

$$(1-as_0t)q_0=1+as_{-1}tq_1=1+atq_1 \tag{2.84}$$

which we can solve for q_1 this time and insert the solution into the above equation

$$b((1-as_0t)q_0-1)=t^2a(aq_0-\widehat{q_0}) \tag{2.85}$$

Collecting coefficients and simplifying leads us to

$$(b(1-as_0t)-a^2t^2)q_0-(-at^2)\widehat{q_0}=b \tag{2.86}$$

$$(c(1-as_0t)-at^2)q_0-(-t^2)\widehat{q_0}=c \tag{2.87}$$

which gives $f(a,t)=c(1-as_0t)-at^2$ so that

Proposition 6. *When $s_{-1}=s_1=1$ we have*

$$B((c(1-as_0t)-at^2)q_0)=c. \tag{2.88}$$

With two different approaches we obtained slightly different relationships upon applying the operator B.

2.2.3 1-dimensional case conclusion

In this section we were able to find explicit solutions for q_0, q and also B-operator relationships for q and $\widehat{q}$. In higher dimensions this may not be quite possible but we should still be able to find B-operator relationships for the equivalent of q_0 and $\widehat{q_0}$. Here will be a table showing the results of this section. Here is a table showing detailed results for each of the 8 possible cases of random walks restricted to the non-negative axis.

Chapter 3

Two-dimensional Random Walks

3.1 Introduction

The power of the operator notation really starts to show here as it allows us to easily derive functional equations for the two-dimensional problem.

Definition 7. *Let $Q = Q(x_1, x_2, a_1, a_2; t)$ be the generating function enumerating two-dimensional quarter-plane random walks with interactions with the axes. Here the step Laurent polynomial is $S = S(x_1, x_2)$ and will be made explicit if the context requires it. The x's indicate the ending coordinates of each walk while the a's count its interactions with the axes.*

Here the operators E and L are defined as $E = E_1 E_2$ where $E_k = 1 - \overline{x_k}[\overline{x_k}]$ and $L = L_1 L_2$ where $L_k = 1 + (a_k - 1)[x_k^0]$. As before we start with the empty walk with monomial 1. Then to find Q we take SQ to obtain all possible extensions of walks enumerated by Q. Then we need to remove all walks stepping outside of the quarter plane. Such walks have either x_1 or x_2 to a negative power (or both) so if we take ESQ we remove all problematic walks. Now we need to mark those walks stepping on the axes so we calculate $LESQ$. Since all the walks enumerated here has one more step we need to keep track of that by multiplying with t. The only missing walk is the empty walk. This yields $Q = 1 + tLESQ$ or in operator form

$$(1 - tLES)Q = 1 \tag{3.1}$$

But we know that the operator L is invertible with inverse $L^{-1} = L_1^{-1}L_2^{-1}$ where $L_k^{-1} = 1 - c_k[x_k^0]$. We can multiply by this inverse to find

$$(L^{-1} - tES)Q = L^{-1}1 = \bar{a}_1\bar{a}_2 = \alpha. \tag{3.2}$$

Proposition 8. *The function Q satisfies the following functional equations*

$$(1 - tLES)Q = 1 \tag{3.3}$$
$$(L^{-1} - tES)Q = \alpha \tag{3.4}$$

We shall expand the equation to its explicit form. We have

$$L^{-1}Q = (1 - c_1[x_1^0])(1 - c_2[x_2^0])Q = Q - c_1Q_1^0 - c_2Q_2^0 + c_1c_2Q_{1,2}^{0,0} \tag{3.5}$$

and

$$ESQ = (1 - \bar{x}_1[\bar{x}_1])(1 - \bar{x}_2[\bar{x}_2])SQ \tag{3.6}$$
$$= SQ - \bar{x}_1S_1^{-1}Q_1^0 - \bar{x}_2S_2^{-1}Q_2^0 + \bar{x}_1\bar{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{0,0} \tag{3.7}$$

We skipped some steps in the latter calculation. It may seem like there should be more terms on the RHS but those terms are zero since Q only contains non-negative powers of x_1 and x_2. Combining all the terms calculated above yields

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

Definition 9. *For $I \subseteq [d]$ let*

$$K_I = c_I - t\bar{x}_IS_I^{-1} \tag{3.8}$$

where if $I = \emptyset$, $K_I = K$.

This allows us to re-write the equation neatly.

Corollary 10. *The function Q satisfies the functional equation*

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \bar{a}_1\bar{a}_2 \tag{3.9}$$

We can also perform coefficient extractions like in the one-dimensional case and obtain further functional equations.

Corollary 11. *The function Q satisfies the additional functional equations*

$$(1-ta_1S_1^0)Q_1^0-(c_2-ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0}-a_1t(S_1^{-1}Q_1^1-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0})=\overline{a_2}$$
$$(1-ta_2S_2^0)Q_2^0-(c_1-ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0}-a_2t(S_2^{-1}Q_2^1-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1})=\overline{a_1}$$
$$(1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0}-ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{0,-1}Q_{1,2}^{0,1})=1$$

where $S_{1,2}^{0,0}=1$ *if we have resting steps.*

We should contrast this quick analysis to the one done without the operator notation. We would have to go through an inclusion-exclusion process that can be rather complicated and prone to error. It doesn't work out too badly when dealing with walks without interactions. However, once we consider the interactions the complexity of the process blows up. This is due to having the *LE* operator together. We would obtain a large equation. Then we wouldn't think about inverting operators because we don't see any. The only visible way to go is through coefficient extraction of the equation. Then we may notice that we can backsubstitute terms and end up with the equation above in terms of K_I's. This process would take several pages and doesn't generalize nicely. However the operator notation allows us to skip the back-substitution step entirely by just inverting L and separating the L and E operators which reduces the number of terms. The case for general dimension d follows nicely and we write it out at a later point.

3.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$

In this work we'll focus on investigating the properties of the function $G(a_1,a_2)=Q_{1,2}^{0,0}(a_1,a_2)$. Let $G_1=G(1,a_2)$, $G_2=G(a_1,1)$ and $G_{1,2}=G(1,1)$. We could also see these as coefficient extractions of G with respect to b_1, b_2 and b_1b_2, i.e. $G_{1,2}=[b_1^0b_2^0]G$. In previous works it has been shown that there is a relationship between these quantities depending on the particular step set S. In some cases (like diagonal walks $S=(x_1+\overline{x_1})(x_2+\overline{x_2})$) we have a linear relationship between these quantities. In other cases (like cartesian walks $S=x_1+\overline{x}_1+x_2+\overline{x}_2$) we don't have such a linear relationship but we have a rational one, that is G is a rational function

of G_1, G_2 and $G_{1,2}$. The aim of the followin sections is to investigate when such relationships exist, when se have a relationship of that type and whether we have other cases.

The work can be split into several cases - walks that never leave the origin and ones that leave but never return to the origin. In the first case the relationship is trivially $G - G_1 - G_2 + G_{1,2} = 0$ because $Q = 1$ itself and so $G = 1$. In the second case Q may be more complicated but again $G = 1$ trivially since no walks end at the origin. This case satisfies the same relationship. However, once we move beyond this stage we need to consider the problem more carefully.

In any of the simple cases we can write $BG = 0$ and sometime $BQ = 0$ where $B = B_1B_2$ from before. We'll investigate the when we'll have a relationship like $B(FQ) = 0$, $B(FG) = 0$ or $B(FG)$ is constant for some F's.

3.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning

Here are all the easy cases of step sets that lead to walks that either don't leave the origin or leave it and never come back. We're not allowing resting steps here. There are 80 such sets among the 256 non-resting possible cases. Of course we could use symmetries to eliminate cases but for now this was an easier thing to do.

There are $2^5 = 32$ types of walks that don't leave the origin. Those are walks with steps among the set {NW,W,SW,S,SE}. Next we look at the walks that leave the origin but never come back. There is the case of $S = \{NE\}$ that never comes back to any axis. Let's look at those walks that never come back to the x_1 axis. We have some steps that go in the positive x_2 direction, that is one of {N, NE, E} together with some steps from {NW, W, SW}. If we were to include any other steps we would come back to the x_1 axis. If we only have a N step, we have 8 new cases, if we have only a NE step we have 3 new cases and if we have both, we have 4 new cases. This gives a total of 15 new cases. There's an equivalent number of walks that leave the x_2 axis and never come back to it. The running total is $32 + 1 + 15 + 15 = 63$.

To finish off we have 17 cases of walks that leave the origin, eventually interact with both axes but never come back to the origin. For this we must only include

steps in such a way that we cannot pick a selection of steps that will decrease both the x_1 and x_2 coordinates. We have to pick a subset of steps from {N, NE, E} so the only other steps allowed are NW and SE. The new cases coming from {N, NE, E}, {N,E}, {N, NE }, {NE, E}, {N}, {NE}, {E} are 4, 4, 2, 2, 1, 2, 2 respectively.

Before we consider walks that come back to the origin we'll look at the behavior Q depending on whether S factors in some nice way or not.

3.4 Simple case: Factoring S into a product

We take a moment to mention a result that falls out of the operator notation and which was covered in more generality by Steve et al. We have a special case when S factors as $S = S_1(x_1)S_2(x_2)$. We have

$$(DS)^k = (D_1D_2S_1S_2)^k = (D_1S_1)^k(D_2S_2)^k \tag{3.10}$$

since the D_1 and D_2 operators commute and the D_iS_i operators also commute since they deal with different variables. The functional equation for Q is

$$(1 - tDS)Q = 1 \tag{3.11}$$

We can compute Q directly via the iteration as

$$Q = \sum_{k=0}^{\infty} t^k(DS)^k = \sum_{k=0}^{\infty} t^k(D_1S_1)^k(D_2S_2)^k = Q_1(x_1, a_1, t) \odot_t Q_2(x_2, a_2, t) \tag{3.12}$$

where $\odot_t$ denotes the Hadamard product of two functions with respect to variable t. From now on we'll drop the subscript t unless it is needed. Here Q_i is given by

$$Q_i = \sum_{k=0}^{\infty} t^k(D_iS_i)^k \tag{3.13}$$

and by reversing the iteration we obtain the functional equation

$$(1 - tD_iS_i)Q_i = 1 \tag{3.14}$$

In other words, $Q_i(x_i, a_i, t)$ is the generating function for a 1-dimensional problem with Laurent step polynomial $S_i(x_i)$. In the case of a factoring of this sort, we are able to glue the 2-dimensional solution out of two 1-dimensional solutions. (Of course this generalizes easily to many dimensions and gives an infinite family of solutions to higher-dimensional problems where S factors into lower dimensional S_i's). If we want to look at the generating function for quarter plane walks of length k it is enough to look at the product of two generating functions for one-dimensional walks of length k. Also, we can write

$$Q = Q_1 \odot Q_2, \quad \widehat{Q} = \widehat{Q}_1 \odot Q_2 \tag{3.15}$$

$$\widehat{Q} = Q_1 \odot \widehat{Q}_2, \quad \widehat{\widehat{Q}} = \widehat{Q}_1 \odot \widehat{Q}_2 \tag{3.16}$$

Therefore

$$B_1 B_2 Q = Q - \widehat{Q} - \widehat{Q} + \widehat{\widehat{Q}} = (Q_1 - \widehat{Q}_1) \odot (Q_2 - \widehat{Q}_2) \tag{3.17}$$

$$B_1 B_2 (Q_1 \odot Q_2) = (B_1 Q_1) \odot (B_2 Q_2) \tag{3.18}$$

In general if we are applying operators that don't involve t to a Hadamard product with respect to t we can treat the operator as a constant with respect to t and apply it on whichever term we would like to. If it applies to both terms then we have to apply it to their mixture in a non-trivial way. For example, each of the Q_i's solves a 1-dimensional problem and therefore we have a relationship of the type $B_i(f_i Q_i) = g_i$ for some f_i and g_i. Then we can write

$$B_1 B_2 (f_1 Q_1 \odot f_2 Q_2) = (B_1 f_1 Q_1) \odot (B_2 f_2 Q_2) = g_1 \odot g_2 \tag{3.19}$$

but so far this is not a relationship of the type $B_1 B_2 (fQ) = g$ since taking out the functions f_1 and f_2 from within the Hadamard product is non-trivial in general. One might be tempted to say that $f = f_1 \odot f_2$ but that would be incorrect. It would be neat to see if there is a way to find f in terms of f_1 and f_2.

There is a way to see the Hadamard product solution without the iteration method. Suppose that Q_i is a solution to the functional equation $(M_i - tE_i S_i)Q_i = \overline{a}_i$ for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional

equation $(M - tES)Q = \overline{a}_1\overline{a}_2$. To this end we calculate

$$(M - tES)Q = M(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) \tag{3.20}$$

$$= (M_1Q_1 \odot M_2Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2)) \tag{3.21}$$

$$\tag{3.22}$$

Then we substitute $M_iQ_i = \overline{a}_i + tE_iS_iQ_i$ into the first term to obtain

$$(M_1Q_1 \odot M_2Q_2) = (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) \tag{3.23}$$

$$= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) + (tE_1S_1Q_1) \odot (tE_2S_2Q_2)) \tag{3.24}$$

We have

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0 \tag{3.25}$$

since we are dealing with power series in t. Therefore

$$(M_1Q_1 \odot M_2Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2)) \tag{3.26}$$

and upon substitution we are left with $\overline{a}_1\overline{a}_2$ which is what we needed in the first place.

One may also ask whether we can do something in the case where S factors and the terms share variables.

This factoring also generalizes to higher dimensions. If S factors into a product of k factors $S_1, \ldots, S_k$ where no two of which share any variables then the problem of finding Q boils down to solving k lower-dimensional problems for functions $Q_1, \ldots, Q_k$, naming variables appropriately and taking the Hadamard product of the functions. That is

$$Q = Q_1 \odot \cdots \odot Q_k \tag{3.27}$$

The more lower-dimensional problems that we solve, the more higher dimensional problems we can solve. Although it appears that the interesting higher dimensional

problems that can be solved in this manner will have a lower and lower proportion among all higher dimensional problems.

3.5 Kernel Group Considerations

It is important to find more functional equations if we can. One way to do this is by applying a transformation on the pair (x_1, x_2) that leaves as many parts of the functional equation as we can but still adding some information in the form of a new equation. One way to do this is to cosider mapping x_1 and x_2 to some quantities separately and then consider the full group generated by such maps. Since we usually eliminate the kernel in front of Q it is useful to consider transformations that preserve the kernel. We'll call this group $G(K)$, although it won't be the full group of rational transformations preserving the kernel. For example $(x_1, x_2) \mapsto (x_2, x_1)$ will not be in the group. Any transformation on (x_1, x_2) that preserves the kernel $K = 1 - tS$ must necessarily preserve the step polynomial S. Here

$$S = \alpha\bar{x} + \beta + \gamma x \tag{3.28}$$

where $x = x_1$ or x_2 and α, β, γ change correspondingly. We consider sending $x \mapsto X$ and solve for X to determine the possible transformations we can perform. That is we have

$$\alpha\bar{x} + \gamma x = \alpha\overline{X} + \gamma X \tag{3.29}$$

$$\alpha(\bar{x} - \overline{X}) = \gamma(X - x) \tag{3.30}$$

$$\alpha\frac{X - x}{Xx} = \gamma(X - x) \tag{3.31}$$

We have cases

- $\alpha = 0, \gamma = 0$. The equation is then trivially true and any transformation preserves the kernel although this is not very useful though.
- $\alpha = 0, \gamma \neq 0$. Then $X = x$ and only the identity transformation preserves the kernel.
- $\alpha \neq 0, \gamma = 0$. Here $X = x$ as well.

- $\alpha \neq 0, \gamma \neq 0$. Here we either have $X = x$ or $X \neq x$ and $X = \bar{x}\alpha/\gamma = \bar{x}S_{-1}/S_1$

In conclusion, either every transformation works and we don't have an interesting case or we have the identity transformation plus a possible second one. The last transformation creates a non-trivial group and we can have groups of various orders. Let

$$g_1 : x_1 \mapsto \overline{x_1}\frac{S_1^{-1}}{S_1^1}, \quad g_2 : x_2 \mapsto \overline{x_2}\frac{S_2^{-1}}{S_2^1} \tag{3.32}$$

whenever the S quantities are non-zero. Note that $g_i(g_i(x_1)) = x_1$, that is they are involutions. We have three cases for each x

- 1: Any transformation can be applied
- 2: We only have the identity transformation
- 3: We have two transformations

We'll combine cases for analysis in the form "case $i.j$" to mean we have case i for x_1 and case j for x_2.

Case 1.1: Here we can apply any transformation on both x_1 and x_2. From the transformation analysis this means that $S_1^{-1} = S_1^1 = S_2^{-1} = S_2^1 = 0$ and thus means that we have no steps that go away from the origin. Therefore $S = S_{1,2}^{0,0}$. Thus we have only two models - no steps or a resting step. (I should include little diagrams with arrows like before).

Case 1.2 : Here any transformation applies on x_1, so $S_1^{-1} = S_1^1 = 0$ so the model has no steps that step away from the origin in the x_1-direction. Since the only transformation on x_2 is the identity transformation we either have $S_2^{-1} = 0$ or $S_2^1 = 0$ in which case the model consists of only an up step or a down step, together with a possible resting step. Similarly, the 2.1 case involves a single x_1 or $\overline{x_1}$ step with a possible resting step.

Case 1.3: (and 3.1): This case is similar to the previous one except that now we have two transformations for x_2 and this means $S_2^{-1}, S_2^1 \neq 0$. This means we must have either both vertical steps x_2 and $\bar{x}_2$ in the 1.3 case or two horizontal steps x_1 and $\bar{x}_1$ in the 3.1 case.

Case 2.2: In this case both transformations are the identity transformations. Therefore exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. WLOG, assume that $S_1^1 \neq 0, S_2^1 \neq 0$ (the other models can be obtained by a rotation around the origing of 90, 180 and 270 degrees). The possible models are (I should include the drawings): {N,E,NE}, {N,NE}, {E,NE}, {N,E}, {NE}. In other words, the subsets of {N,E,NE} except the ones going either only in the N direction or only in the E direction. This is a case of group $G(K)$ or order 1.

Case 2.3: In this case there is only one transformation for x_1 and two for x_2. This means $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case). Also $S_2^{-1}, S_2^1 \neq 0$ so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. We have 17 cases (and another 17 including a resting step). Of course in the 3.2 case we have the meaning of x_1 and x_2 reversed.This is a case of group $G(K)$ of order 2.

Case 3.3: This is the most interesting case with two transformations for x_1 and x_2 each. This means that there must be at least one step going in a positive and one going in a negative coordinate for each of x_1 and x_2. There are a lot of such cases (in the hundreds). Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity depending on the step set in question.

Later we'll see that these cases correspond to the idea of dimensionality of the model. Case 1.1 is zero-dimensional. Cases 1.2, 1.3 (and 3.1) are one-dimensional. Cases 2.2, 2.3 and 3.3 are two-dimensional. It should not be a suprise that the 0 and 1-dimensional cases will be easy while the two-dimensional cases will become progressively more difficult.

3.6 Solving basic 2D models

3.6.1 Cases 1.1, 1.2, 1.3

A quick glance at the step models involved shows that we are dealing with essentially one-dimensional problems. This means they can be solved easily explicitly.

One can also note that these models are all factorable

$$S(x_1,x_2) = 1 \cdot S(x_2) \tag{3.33}$$

Both Q_1 and Q_2 are functions we solved for in the one-dimensional case

$$Q_1(x_1,a_1,t) = \frac{1}{1-a_1 t}, \quad Q_2(x_2,a_2,t) = q(x_2,a_2,t) \tag{3.34}$$

for some solution q from that section. Therefore

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = \frac{1}{1-a_1 t} \odot q(x_2,a_2,t) = q(x_2,a_2,a_1 t) \tag{3.35}$$

Similarly in the 2.1, 3.1 cases

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = q(x_1,a_1,t) \odot \frac{1}{1-a_2 t} = q(x_1,a_1,a_2 t) \tag{3.36}$$

and this completes the solution. What about the relationship with the $B_1 B_2$ transformation? If $f(a,t)$ is the function that goes together with $q(x,a,t)$ then the function $F(a_1,a_2,t)$ that goes together with $Q(x_1,x_2,a_1,a_2,t)$ is given by

$$F(a_1,a_2,t) = f(a_2,a_1 t) \tag{3.37}$$

Then we have $B_2(FQ) = 0$ and $B_1 B_2(FQ) = 0$.

3.6.2 Case 2.2 - group of order 1

A quick glance at the step models reveals that this case is mostly trivial. We have a lot of models that never leave the origin and those have a trivial solution

$$Q(x_1,x_2,a_1,a_2) = \frac{1}{1-a_1 a_2 S_{1,2}^{0,0} t}. \tag{3.38}$$

We also have a lot of models that are essentially one-dimensional with a single step going along an axis and some steps going off-axis where the latter steps can

never be taken by a random walk in this model. They are essentially cases already covered in the previous section. Suppose that the single step is taken in the positive x_1 direction. This is equivalent to the $s_{-1} = 0$ case in 1D with function

$$q(x,a,t) = \frac{1 - s_0 t}{(1 - st)(1 - as_0 t)} = \frac{1 - s_0 t}{(1 - x_1 s_1 t)(1 - as_0 t)} \tag{3.39}$$

Then we have

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = q(x_1,a_1,t) \odot \frac{1}{1 - a_2 t} = q(x_1,a_1,a_2 t) \tag{3.40}$$

and therefore

$$Q(x_1,x_2,a_1,a_2,t) = \frac{1 - a_2 S^{0,0}_{1,2} t}{(1 - (x_1 + S^{0,0}_{1,2}) a_2 t)(1 - a_1 a_2 S^{0,0}_{1,2} t)} \tag{3.41}$$

The interesting models in case 2.2 are those with non-negative steps that we listed. Here

$$S = \alpha x_1 + \beta x_2 + \gamma x_1 x_2 + \delta \tag{3.42}$$

The functional equation and its zero extractions are

$$(1 - tS)Q - c_1 Q^0_1 - c_2 Q^0_2 + c_1 c_2 Q^{0,0}_{1,2} = \overline{a_1 a_2} \tag{3.43}$$

$$(1 - t a_1 (\delta + \beta x_2)) Q^0_1 - c_2 Q^{0,0}_{1,2} = \overline{a_2} \tag{3.44}$$

$$(1 - t a_2 (\delta + \alpha x_1)) Q^0_2 - c_1 Q^{0,0}_{1,2} = \overline{a_1} \tag{3.45}$$

$$(1 - a_1 a_2 t \delta) Q^{0,0}_{1,2} = 1 \tag{3.46}$$

We solve the problem by a simple back-substitution of the variables.

$$Q_{1,2}^{0,0} = \frac{1}{1-a_1a_2\delta t} \tag{3.47}$$

$$Q_1^0 = \frac{\overline{a}_2}{1-ta_2S_1^0}\left(1+\frac{b_2}{1-a_1a_2\delta t}\right) = \frac{1-a_1\delta t}{(1-ta_1S_1^0)(1-a_1a_2\delta t)} \tag{3.48}$$

$$Q_2^0 = \frac{\overline{a}_1}{1-ta_1S_2^0}\left(1+\frac{b_1}{1-a_1a_2\delta t}\right) = \frac{1-a_2\delta t}{(1-ta_2S_2^0)(1-a_1a_2\delta t)} \tag{3.49}$$

$$Q = \frac{1}{1-tS}(\overline{a}_1\overline{a}_2 + c_1Q_1^0 + c_2Q_2^0 - c_1c_2Q_{1,2}^{0,0}) \tag{3.50}$$

$$= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-ta_1a_2\delta}\left(c_1\frac{1-a_1\delta t}{1-ta_1S_1^0} + c_2\frac{1-a_2\delta t}{1-ta_2S_1^0} - c_1c_2\right)\right) \tag{3.51}$$

$$= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-a_1a_2\delta t}(c_1f_1(a_1,t) + c_2f_2(a_2,t) - c_1c_2)\right) \tag{3.52}$$

where the last line explicitly shows that Q is composed of functions that depend only on one of a_1, a_2. If we want to find a relation of the type $B(FQ) = 0$ then we must first find a suitable function F. First of all the $1-tS$ part doesn't play a role since it doesn't depend on a_1 or a_2. Then later the variables c_1 and c_2 will vanish when extracting the coefficients of b_1^0 and b_2^0. We'll now show that the function

$$F(a_1,a_2,t) = 1-a_1a_2\delta t \tag{3.53}$$

works exactly. First

$$B((1-a_1a_2\delta t)\overline{a}_1\overline{a_2}) = B(\overline{a}_1\overline{a}_2 - t\delta) = B(\overline{a}_1\overline{a}_2) - B(t\delta) = \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1c_2 \tag{3.54}$$

Then

$$B(c_1f_1 + c_2f_2 - c_1c_2) = c_1f_1 + c_2f_2 - c_1c_2 - c_2f_2 - c_1f_1 = -c_2c_2 \tag{3.55}$$

Putting the two quantities together along with dividing by $1-tS$ shows that $B(FQ) = 0$. Of course this works for $G = Q_{1,2}^{0,0}$ too.

3.6.3 Case 2.3 - group of order 2

There are two possibilities here - positive and negative. In the negative case $S_1^1 = 0$ and the steps have a non-positive x_1 component. Due to the fact that a lot of steps are not going to be taken by random walks in those models, all of these models reduce to the cases of S consising of a rest step and maybe a single step in a positive direction. This takes care of 34 models (or 68 if we include a rest step). The interesting models are those with a positive x_1 component, or in other words, models where $S_1^{-1} = 0$. The calculations for these models become increasingly more complicated. The functional equation and its extracted forms read as

$$(1-tS)Q - c_1Q_1^0 - (c_2 - t\bar{x}_2S_2^{-1})Q_2^0 + c_1c_2Q_{1,2}^{0,0} = \overline{a_1a_2} \tag{3.56}$$

$$(1-ta_1S_1^0)Q_1^0 - (c_2 - ta_1\bar{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} = \overline{a_2} \tag{3.57}$$

$$(1-ta_2S_2^0)Q_2^0 - c_1Q_{1,2}^{0,0} - a_2tS_2^{-1}Q_2^1 = \overline{a_1} \tag{3.58}$$

$$(1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2S_{1,2}^{0,-1}Q_{1,2}^{0,1} = 1 \tag{3.59}$$

This is starting to look somewhat daunting. However, we can identify a subproblem that is essentially one-dimensional and therefore already solved. For this we look at the second and fourth equations with the substitution $a_i = a, c_i = c, x_2 = x, ta_1 = \tau, S_1^0 = T, Q_1^0(x_2) = q(x)$. These equations become

$$(1-\tau T)q - (c - \tau\bar{x}T_{-1})q_0 = \overline{a} \tag{3.60}$$

$$(1-a\tau T_0)q_0 - a\tau S_{1,2}^{0,-1}q_1 = 1 \tag{3.61}$$

where $q_0 = q(a,\tau)$. This is the one-dimensional problem again. The solution here is $q(x,a,\tau)$ so the original problem has a solution

$$Q_1^0(x_2) = q(x_2,a_2,a_1t), \quad Q_{1,2}^{0,0} = q_0(a_2,a_1t). \tag{3.62}$$

The reason the method worked in this case is that we have steps that keep a random walk staying on the x_2 axis and steps that take it away from the x_2 but no steps that have it return to it. This is similar to the case where a random walk leaves the x_2 axis never to come back. Essentially the problem splits into a one-dimensional one regarding the x_2 axis and a further 2D one once we step off the axis. We re-write

the functional equation as

$$(1-tS)Q-(c_2-t\bar{x}_2S_2^{-1})Q_2^0=\bar{a}_1\phi(x_2)=\overline{a_1}(\bar{a}_2+b_1Q_1^0-b_1c_2Q_{1,2}^{0,0}) \qquad (3.63)$$

Here we can treat x_2 as a constant and solve for a one-dimensional problem with x_1 as a variable. We can perform the same analysis as in the the one-dimensional case except where we had $\bar{a}_1$ we would now have $\bar{a}_1\phi(x_2)$. Therefore we can solve for Q explicitly. However I haven't done this yet so I'll leave this for later. Curiously enough this analysis does not use the fact that we are dealing with a group of order 2 anymore than the one-dimensional case uses the fact that $|G(K)|=2$ for $K=1-t(\bar{x}+x)$. Interestingly we can try re-doing the problem using the group structure. It would surely illustrate the methods we would use later in analysing the higher group size models. I've done it on paper and I don't think I've made any mistakes. However I don't obtain the correct answers. I suspect it's because of the following - $\sigma_1=1/(tS_1^1)-S_1^0/S_1^1$ which does not expand as a power series in t. Perhaps we just cannot plug this solution into the functional equations. Therefore one would have to resort to using the σ_2 solution without being able to use the group structure at all. I can certainly solve the problem in the above manner and use the series generated for each part of the solution to test whether the second solution really fails, and if it does test if it fails because of the σ_1 behavior.

Chapter 4

Hexagonal Walks

4.1 Introduction

We'll tackle the problem of random walks on a honey-comb lattice. This is an example of a problem set on a different topology. It's our first problem dealing with a random walk model where the direction in which you go depends on where you are on the lattice. It can be interpreted as a random walk model on the plane where we have two distinct step polynomials depending on where we are. Because of this the flavour of the problem will be quite different. However it shall serve as a good example of the use of operators. We start by embedding the honeycomb lattice in $\mathbb{Z}^2$ where the first hexagon has coordinates

$$\{(0,0),(1,1),(1,2),(0,0),(-1,2),(-1,1)\} \tag{4.1}$$

and the rest of the lattice is obtained by tiling the plane with this hexagon. As before we let Q be the generating function for the random walks on the lattice where x, y keep track of the endpoint of a walk and t keeps track of its length. Of course, off the lattice the terms will have zero coefficients. Out of thin air we create two operators C_1 and C_2, acting on Q, which pick out all the bottom and top corners of the hexagon tiles respectively. By picking out we mean that C_1 results in the partial generating function corresponding the the coordinates given in the two situations. The condition is that $C_1 + C_2 = 1$ the identity. These two locations give two distinct

possibilities for step sets so we'll give them their own step polynomials. Let's call their respective step polynomials S_1 and S_2 given by

$$S_1 = (x+\overline{x})y+\overline{y}, \quad S_2 = (x+\overline{x})\overline{y}+y \tag{4.2}$$

Now we can continue to create a functional equation for the hexagonal model. We perform an iteration. We start with the empty walk with generating function $Q = 1$. Then for every iterate of Q we create the next iterate by taking the walks enumerated in Q and extending them by one step. Since we have two situations we need to extend by different steps in each case. That is we start with Q, extract the S_1-centric walks with C_1Q and then extend them by multiplying with S_1. The result is that the new Q is given by

$$Q = 1+t(S_1C_1+S_2C_2)Q = 1+t\Sigma Q. \tag{4.3}$$

We can also write this in the operator format as

$$(1-t\Sigma)Q = 1 \tag{4.4}$$

We may wish to restrict the random walks to the quarter plane or in some other way. To do this we need to also use an extraction operator E that removes those walks that stray out of the boundaries. Since each step is of size at most 1 we only need to remove those walks that step 1 unit outside of the boundaries and E is then easy to write. The iteration and functional equation are given by

$$Q = 1+tE(S_1C_1+S_2C_2)Q = 1+tE\Sigma Q \tag{4.5}$$

$$(1-tE\Sigma)Q = 1 \tag{4.6}$$

Of course we can also label the walks that interact with specific locations. We repeate the work with the extra operator L. In general we can use an operator Γ to encapsulate everything extra that we do. It remains to find the actual operators C_1 and C_2.

4.2 Operators

Before we find the operators we need to find which points they need to act on. We can find all the lower vertices by starting with the hexagon with lower vertex $(0,0)$ and translating it along the lattice. It's not difficult to see that the lower vertices of these hexagons are those with coordinates (m,n) where

- $m \equiv 0 \pmod 2, n \equiv 0 \pmod 4$
- $m \equiv 1 \pmod 2, n \equiv 2 \pmod 4$

The upper vertices of these hexagons are the above coordinates translated by $(0,3)$ in the vertical direction. Those are the coordinates (m,n) where

- $m \equiv 0 \pmod 2, n \equiv 3 \pmod 4$
- $m \equiv 1 \pmod 2, n \equiv 1 \pmod 4$

This allows us to write the operators as

$$C_1 = \sum_{m,n} x^{2m} y^{4n} [x^{2m} y^{4n}] + \sum_{m,n} x^{2m+1} y^{4n+2} [x^{2m+1} y^{4n+2}] \tag{4.7}$$

$$C_2 = \sum_{m,n} x^{2m+1} y^{4n+1} [x^{2m+1} y^{4n+1}] + \sum_{m,n} x^{2m} y^{4n+3} [x^{2m} y^{4n+3}] \tag{4.8}$$

$$\tag{4.9}$$

We can write these operators more succinctly by using the following conventions

$$d_{j,k}(z) = \sum_{m} z^{jm+k} [z^{jm+k}] \tag{4.10}$$

$$D_{ij} = d_{2,i}(x) d_{4,j}(y) \tag{4.11}$$

$$D = D_{00} \tag{4.12}$$

Note that

$$D_{ij} = x^i y^j D \bar{x}^i \bar{y}^j \tag{4.13}$$

This allows us to write the operators C_1 and C_2 succinctly as

$$C_1 = D_{00} + D_{12} = D + xy^2 D\overline{xy}^2 \tag{4.14}$$

$$C_2 = D_{03} + D_{11} = y^3 D\overline{y}^3 + xyD\overline{xy} \tag{4.15}$$

The functional equation now reads

$$(1 - t(S_1(D_{00} + D_{12}) + S_2(D_{03} + D_{11}))Q = 1 \tag{4.16}$$

and analogously we can append an operator Γ to to the expression.

4.3 Group Structure

There are nicer ways to express these extractions without performing the extractions themselves. Let k be a positive integer and $w_k = e^{2\pi i/k}$. Then it is a somewhat well-known result that we can extract the coefficient $0 \pmod k$ of $f(x)$ with the following summation

$$d_{k,0}(x)f(x) = \frac{1}{k}\sum_{j=0}^{k-1} f(w_k^j x). \tag{4.17}$$

This establishes the concrete operator representations

$$d_{2,0}(x)f(x) = \frac{1}{2}(f(x) + f(-x)) \tag{4.18}$$

$$d_{4,0}(y)f(y) = \frac{1}{4}(f(y) + f(iy) + f(-y) + f(-iy)) \tag{4.19}$$

Using this definition in the functional equation would lead to a functional equation relating $Q(x,y)$ to functions Q evaluated at different arguments like $(-x,y)$, (x,iy), etc. This allows us to explore the group structure of the functional equation. To this end let $g : x \mapsto -x$ and $h : y \mapsto iy$ be the two commuting operations and the group of symmetries be

$$G \cong \langle g \rangle \times \langle h \rangle \cong \langle -1 \rangle \times \langle i \rangle \tag{4.20}$$

It remains to see how these group actions affect the various quantities in the problem. First we deal with S_1 and S_2 and then with the operators D_{jk}. This will allow us to find how the actions affect Σ and the functional equation as a whole. First we have the actions of g

$$gS_1 = S_1^* = -(x+\overline{x})y+\overline{y} \tag{4.21}$$

$$gS_2 = S_2^* = -(x+\overline{x})\overline{y}+y \tag{4.22}$$

Then the actions of h

$$hS_1 = i(x+\overline{x})y - i\overline{y} = -igS_1 = -iS_1^* \tag{4.23}$$

$$hS_2 = -i(x+\overline{x})\overline{y} + iy = igS_2 = iS_2^* \tag{4.24}$$

These relatons allows us to complete the rest of the relations iteratively (maybe a multiplication table would be good here)

$$hS_1 = -iS_1^*, \quad h^2S_1 = -S_1, \quad h^3S_1 = iS_1^* \tag{4.25}$$

$$ghS_1 = -iS_1, \quad gh^2S_1 = -S_1^*, \quad gh^3S_1 = -iS_1 \tag{4.26}$$

$$hS_2 = iS_2^*, \quad h^2S_2 = -S_2, \quad h^3S_2 = iS_2^* \tag{4.27}$$

$$ghS_2 = iS_2, \quad gh^2S_2 = -S_2^*, \quad gh^3S_2 = -iS_2 \tag{4.28}$$

$$\tag{4.29}$$

Now g acts on D_{jk} in the following way

$$g(D_{jk}f(x,y)) = g\sum_{m,n} x^{2m+j}y^{4m+k}[x^{2m+j}y^{4m+k}]f(x,y) \tag{4.30}$$

$$= \sum_{m,n}(-x)^{2m+j}y^{4m+k}f_{2m+j,4m+k} \tag{4.31}$$

$$= (-1)^j D_{jk}f(x,y). \tag{4.32}$$

Similarly h acts on D_{jk} in the following way

$$h(D_jkf(x,y)) = i^k D_{jk}f(x,y). \tag{4.33}$$

Or specifically

$$gD_{00} = D_{00}, \quad gD_{11} = -D_{11}, \quad gD_{03} = D_{03}, \quad gD_{12} = -D_{12} \tag{4.34}$$

$$hD_{00} = D_{00}, \quad hD_{11} = iD_{11}, \quad hD_{03} = -iD_{03}, \quad hD_{12} = -D_{12} \tag{4.35}$$

$$\tag{4.36}$$

We can now move on to finding the actions on the operator Σ. To spare some of the work it is enough to find the actions $1, gh, h^2$ and gh^3 since we'll have only 4 independent relations on the operators. Let

$$A = S_1(D_{00} + D_{12}), \quad B = S_2(D_{11} + D_{03}) \tag{4.37}$$

Then

$$\Sigma = B + A, \quad gh\Sigma = B - iA \tag{4.38}$$

$$h^2\Sigma = B - A, \quad gh^3\Sigma = B + iA \tag{4.39}$$

We have

$$(1 + h^2)\Sigma = gh(1 + h^2)\Sigma = 2B \tag{4.40}$$

and

$$(1 - h^2)\Sigma = igh(1 - h^2) = 2A \tag{4.41}$$

By using these eliminations we can write new forms of the functional equation. We have

$$\frac{1}{2}(1 + h^2)(1 - t\Sigma)Q = \left(\frac{1}{2}(1 + h^2) - tB\right)Q \tag{4.42}$$

and

$$\frac{1}{2}(1 + h^2)(1) = \frac{1}{2}2 = 1 \tag{4.43}$$

so the functional equation is

$$\left(\frac{1}{2}(1+h^2) - tS_2(D_{11}+D_{03})\right) Q = 1 \tag{4.44}$$

Repeating the analysis with the other relations yields

$$\left(\frac{1}{2}(1-h^2) - tS_1(D_{00}+D_{12})\right) Q = 0 \tag{4.45}$$

The relations allow us to write two different representations of Σ

$$2\Sigma = ((1+h^2)+igh(1-h^2))\Sigma = ((1-h^2)+gh(1+h^2))\Sigma \tag{4.46}$$

Or in other words

$$(1-h^2-igh(1-h^2))\Sigma = (1-h^2)(1-igh)\Sigma = 0 \tag{4.47}$$

$$(1+h^2-gh(1+h^2))\Sigma = (1+h^2)(1-gh)\Sigma = 0 \tag{4.48}$$

We can use these on the functional equation to kill Σ

$$(1-h^2)(1-igh)(1-t\Sigma)Q = (1-h^2)(1-igh)Q = 0 = (1-h^2)(1-igh)1 \tag{4.49}$$

$$(1+h^2)(1-gh)(1-t\Sigma)Q = (1+h^2)(1-gh)Q = 0 = (1+h^2)(1-gh)1 \tag{4.50}$$

Finally we can write the two relations as

Proposition 12. *The function Q satisfies the functional equation*

$$(1-h^2)(1-igh)Q = (1+h^2)(1-gh)Q = 0 \tag{4.51}$$

where we need both factors to get zero.

If we use the series definition for Q here we find that these relations are saying nothing more than the fact that the terms of Q corresponding to the ”holes” inside the hexagons are zero.

4.4 Finding the generating function Q

Note that

$$\frac{1}{2}(1+h^2)Q(x,y) = \frac{1}{2}(Q(x,y)+Q(x,-y)) = C_1Q \tag{4.52}$$

$$\frac{1}{2}(1-h^2)Q(x,y) = \frac{1}{2}(Q(x,y)-Q(x,-y)) = C_2Q \tag{4.53}$$

where the latter equalities follow since we are dealing with the even and odd y parts of Q respectively, which cover conditions C_1 and C_2. We can re-write the two functional equations from the previous section as

$$C_1Q - tS_2C_2Q = 1 \tag{4.54}$$

$$C_2Q - tS_1C_1Q = 0 \tag{4.55}$$

This is a linear system in the quantities C_1Q and C_2Q. The associated determinant is $1-t^2S_1S_2 \neq 0$ so we find the unique solution to be

$$C_1Q = \frac{1}{1-t^2S_1S_2} = \frac{x^2y^2}{x^2y^2 - t^2(x+y^2+x^2y^2)(1+x^2+xy^2)} \tag{4.56}$$

$$= 1 + (3+\overline{x}^2+x^2+\overline{x}\overline{y}^2+x\overline{y}^2+y^2\overline{x}+xy^2)t^2+\ldots \tag{4.57}$$

$$C_2Q = \frac{tS_1}{1-t^2S_1S_2} = \frac{txy(x+(1+x^2)y^2)}{x^2y^2 - t^2(x+y^2+x^2y^2)(1+x^2+xy^2)} \tag{4.58}$$

$$= (\overline{y}+y\overline{x}+xy)t+\ldots \tag{4.59}$$

Finally

$$Q = C_1Q + C_2Q = \frac{1+tS_1}{1-t^2S_1S_2} = \sum_{k=0}^{\infty} t^{2k}S_1^kS_2^k + \sum_{k=0}^{\infty} t^{2k+1}S_1^{k+1}S_2^k \tag{4.60}$$

Proposition 13. *The generating function Q for unrestricted random walks on the*

honeycomb lattice is given by

$$Q(x,y;t) = \frac{x^2y^2 + txy(x+y^2+x^2y^2)}{x^2y^2 - t^2(1+x^2+xy^2)(x+y^2+x^2y^2)} \tag{4.61}$$

$$= 1 + (\overline{y} + y\overline{x} + xy)t + (3 + \overline{x}^2 + x^2 + \overline{xy}^2 + x\overline{y}^2 + y^2\overline{x} + xy^2)t^2 + \ldots \tag{4.62}$$

An alternative solution is using the properties of $\Sigma = \Sigma_1 + \Sigma_2$ itself. First note that

$$\Sigma_1^2 = \Sigma_2^2 = 0 \tag{4.63}$$

for any input because the first application of Σ_i takes a function and using C_i outputs monomials that conform to C_i. However then we multiply these by S_i which makes them no longer conform to C_i. The result follows. Therefore Σ^k consists of of a sum of k-letter words on Σ_1 and Σ_2 without consecutive appearances. There are only two such sequences - one starting with Σ_1 and the other with Σ_2. Also $\Sigma_1(1) = S_1$ but $\Sigma_2(1) = 0$. Therefore

$$Q = \sum_{k=0}^{\infty} t^k \Sigma^k(1) = \tag{4.64}$$

$$= \sum_{k=0}^{\infty} t^{2k}((\Sigma_1\Sigma_2)^k(1+t\Sigma_1) + (\Sigma_2\Sigma_1)^k(1+t\Sigma_2))(1) \tag{4.65}$$

$$= \sum_{k=0}^{\infty} t^{2k}((\Sigma_1\Sigma_2)^k t\Sigma_1 + (\Sigma_2\Sigma_1)^k)(1) \tag{4.66}$$

$$\tag{4.67}$$

and through some more manipulation we can get

$$Q = \sum_{k=0}^{\infty} t^{2k}(1+tS_1)(\Sigma_2\Sigma_1)^k(1) = (1+tS_1)\sum_{k=0}^{\infty} t^{2k}(S_1S_2)^k \tag{4.68}$$

and the result follows.

Chapter 5

Higher-dimensional walks

5.1 Using Operators to Derive the Functional Equation

As mentioned previously, the generalization to higher-dimensions is straight-forward. Here we have an operators of the form $D = D_1 \dots D_d$. We take the generating function Q enumerating all random walks from model S with interactions. We extend all walks by one step via the multiplication SQ. Then we remove all walks venturing outside of the positive orthant by applying E, that is we have ESQ. We mark all walks that have ended up on the zero hyperplanes via L. The result is $LESQ$. We need to multiply by t to track the extension of the walks. The resulting expression is almost the same as Q itself. The only walk missing is the empty walk with monomial 1. This allows us to write the functional equation for Q as

$$Q = 1 + tLESQ \tag{5.1}$$

which we can re-write as

$$(1 - tLES)Q = 1 \tag{5.2}$$

We can apply the operator L^{-1} on this to obtain

$$L^{-1}(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]} \tag{5.3}$$

and

$$(L^{-1} - tES)Q = \overline{a}_{[d]} \tag{5.4}$$

This equation will contain many less terms. To see this we take a look at the L_iE_i operator

$$L_iE_i = 1 + b_i[x_i^0] - \overline{x_i}[\overline{x_i}] \tag{5.5}$$

which increases the number of terms in the equation by a factor of 3. That is we would have 3^d terms relating to SQ by the end of applying LE. Not to mention that each extraction from SQ introduces 2 terms. On the other hand once we apply the operator L^{-1} we introduce 2^d terms relating to Q but then since we only have E applied on SQ we'd only have at most 2^d terms from this. For the case of walks of step size at most 1 and labelling the hyperplane interactions we have exactly 2^d terms $Q_I^{\mathbb{O}}$, one for each subset $I \subseteq [d]$. In this case we can write out the functional equation explicitly. Firstly,

$$\begin{aligned} L^{-1}Q &= (1 - c_1[x_1^0]) \dots (1 - c_d[x_d^0])Q && (5.6)\\ &= \sum_{I \subseteq [d]} (-1)^{|I|} c_I [x_I^{\mathbb{O}}]Q && (5.7)\\ &= \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{O}} && (5.8) \end{aligned}$$

and

$$\begin{aligned} ESQ &= (1 - \overline{x}_1[\overline{x}_1]) \dots (1 - \overline{x}_d[\overline{x_d}])(SQ) && (5.9)\\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [x_I^{-\mathbb{I}}](SQ) && (5.10)\\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} && (5.11) \end{aligned}$$

Therefore the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}} \tag{5.12}$$

As in the two-dimensional case we let $K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$ where if $I = \emptyset$ we have the empty products and empty extractions leaving us with $K_\emptyset = K$. Finally we can write the master equation in d-dimensions.

Theorem 14. *The function Q satisfies the functional equation*

$$\sum_{I\subseteq[d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a}_{[d]} \tag{5.13}$$

where $K_I = (c_I - t\overline{x_I}S_I^{-\mathbb{I}})$.

As in the two-dimensional case we can now perform coefficient extractions with $[x_I^{\mathbb{O}}]$ to obtain further functional equations relating $Q_I^{\mathbb{O}}$ to other Q terms of the same variables or fewer variables.

5.2 Coefficient Extraction

We can perform coefficient extraction like in the 1-dimensional case. The operator $[x_i^k]$ will commute with all M_j, E_j with $j \neq i$ so that we only need to consider the action on the operators $M_i = L_i^{-1}$ and E_i. We've already done so in the 1-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a_i}[x_i^0] \tag{5.14}$$

$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x_i}[\overline{x_i}] = [x_i^0] \tag{5.15}$$

and the k-th power extraction for $k \geq 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k] \tag{5.16}$$

$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x_i}[\overline{x_i}] = [x_i^k] \tag{5.17}$$

So, for $k \geq 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture while for $k = 0$ it removes E_i but leaves a residual $\overline{a_i}$ behind for M_i. Therefore if we extract

$[x_I^{\mathbb{O}}]$ from the functional equation we have

$$[x_I^{\mathbb{O}}]a_{[d]} = \overline{a}_{[d]} = [x_I^{\mathbb{O}}](M_{[d]} - tE_{[d]}S)Q \tag{5.18}$$

$$= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{O}}] - tE_{[d]\setminus I}[x_I^{\mathbb{O}}]S\right) Q \tag{5.19}$$

$$= [x_I^{\mathbb{O}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q \tag{5.20}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{O}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q \tag{5.21}$$

Written in another way this becomes

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{O}}](M_I - a_{[d]\setminus I}tE_I S)Q \tag{5.22}$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that when we extract coefficients for walks that end at some hyperplane intersection we are having a reduced version of the functional equation at the cost of an added a term. When we extract a general term we can use the operator $[x_I^{\mathbb{O}} x_J^P]$ where P is a set of powers greater than 0 and $I \cap J = \emptyset$, $|P| \geq 1$ and $I \cup J = K$. We find

$$[x_I^{\mathbb{O}} x_J^P]a_{[d]} = 0 = [x_I^{\mathbb{O}} x_J^P](M_{[d]} - tE_{[d]}S)Q \tag{5.23}$$

$$= [x_I^{\mathbb{O}} x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q \tag{5.24}$$

which we can write as

$$[x_I^{\mathbb{O}} x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0 \tag{5.25}$$

If I is empty then

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0 \tag{5.26}$$

The problem with a lot of things in this section is that they are not really practical beyond providing a shortcut for some of the calculations so we don't have to do

them by hand every time. Also they provide a broader idea of having multiple kernels in one problem. Perhaps things will be clearer when we specialize to the 2-dimensional problem.

5.3 Further functional equations

We can try going a little further in the coefficient extraction to come up with an equation that looks like the original functional equation. We know that

$$[x_I^{\mathbb{O}}]SQ = S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I \tag{5.27}$$

where R_I is a collection of $S_I^{-P} Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this inside the functional equation

$$\begin{aligned}
\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{O}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{O}}](SQ) && (5.28)\\
&= M_{[d]\setminus I} Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I) && (5.29)\\
&= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I} R_I && (5.30)
\end{aligned}$$

so we can re-write the functional equation as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I') \tag{5.31}$$

where $R_I' = E_{[d]\setminus I} R_I$. We can calculate the following terms

$$M_{[d]\setminus I} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} c_J Q_{I,J}^{\mathbb{O},\mathbb{O}} \tag{5.32}$$

and

$$a_I t E_{[d]\setminus I} S_I^{\mathbb{O}} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} a_I t \overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}} Q_{I,J}^{\mathbb{O},\mathbb{O}} \tag{5.33}$$

so the functional equation can be written as

$$\sum_{J \subseteq [d]\setminus I} (-1)^{|J|} (c_J - a_I t \overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}) Q_{I,J}^{\mathbb{O},\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I') \tag{5.34}$$

This suggests the notation

Definition 15. *For sets I and J we write*

$$K_J(I) = c_J - a_I t \bar{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}} \tag{5.35}$$

where when $I = \emptyset$, $K_J(I) = K_J$.

This notation allows us to write the 2^d functional equations in the simplified form

Corollary 16. *For $I \subseteq [d]$ the function Q satisfies the functional equation*

$$\sum_{J \subseteq [d] \setminus I} (-1)^{|J|} K_J(I) Q_{I,J}^{\mathbb{O},\mathbb{O}} = \overline{a}_{[d]\setminus I} + a_I t R'_I \tag{5.36}$$

where the R'_I term involves only higher order coefficient extractions from Q, that is not all the powers are zero.

Now, to calculate the R'_I terms we first determine what R_I is. It is a sum of terms $S_I^{-P} Q_I^P$ for various nonnegative P with size $|I|$ whose elements are not all 0. To calculate R'_I we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in the remaining variables so they don't have non-negative coefficients. Let $sq = S_I^{-P} Q_I^P$. Then

$$\begin{aligned} E_j sq &= sq - \bar{x}_j (sq)_j^{-1} && (5.37)\\ &= sq - \bar{x}_j (s_j^{-1} q_j^0 + s_j^0 q_j^{-1} + s_j^1 q_j^{-2}) && (5.38)\\ &= sq - \bar{x}_j s_j^{-1} q_j^0 && (5.39) \end{aligned}$$

Hence

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \bar{x}_J s_J^{-\mathbb{I}} q_J^{\mathbb{O}} \tag{5.40}$$

Finally we can write

$$R'_I = E_{[d]\setminus I} R_I \tag{5.41}$$
$$= \sum_{P>\mathbb{0}} E_{[d]\setminus I} S_I^{-P} Q_I^P \tag{5.42}$$
$$= \sum_{P>\mathbb{0}} \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{0}} \tag{5.43}$$
$$= \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} \sum_{P>\mathbb{0}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{0}} \tag{5.44}$$
$$\tag{5.45}$$

Proposition 17. *The functions R'_I are given by*

$$R'_I = \sum_{J\subseteq[d]\setminus I} (-1)^{|J|} \sum_{P>\mathbb{0}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{0}} \tag{5.46}$$
$$\tag{5.47}$$

This formula is mostly of theoretical value although it can be used to speed up writing things out by hand.

5.4 Using the formal inverse for iteration

In a previous subsection we obtained the equation

$$(1 - tLES)Q = 1 \tag{5.48}$$

Let $D = LE$. Then we can think of the formal inverse of this operator against some norm $||tDS||$ or even formally. Formally

$$(1 - tDS)^{-1} = \frac{1}{1 - tDS} = \sum_{k=0}^{\infty} t^k (DS)^k \tag{5.49}$$

so we can write the solution for Q as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k (1) \tag{5.50}$$

where $(DS)^k(1) = DS(\ldots(DS(1))\ldots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with DS applied to it. It can be done fairly efficiently on a computer.

Example 18. Let $S = x_1 + \bar{x}_1 + x_2 + \bar{x}_2 = S_1(x_1) + S_2(x_2)$. We start with $DS^0(1) = 1$. Then

$$DS^1(1) = LE(S_1 + S_2) = L(x_1 + x_2) = a_2x_1 + a_1x_2 \tag{5.51}$$
$$DS^2(1) = LE(S_1 + S_2)(a_2x_1 + a_1x_2) \tag{5.52}$$
$$= LE(a_2S_1x_1 + a_1x_2S_2 + a_2x_1S_2 + a_1x_2S_1) \tag{5.53}$$
$$= L(a_2S_1x_1 + a_1x_2S_2 + (a_1 + a_2)x_1x_2) \tag{5.54}$$
$$= a_1a_2^2 + a_2^2x_1^2 + a_1a_2^2 + a_1^2x_2^2 + (a_1 + a_2)x_1x_2 \tag{5.55}$$

which allows us to write

$$Q = 1 + (a_2x_1 + a_1x_2)t + \left(a_1a_2(a_1 + a_2) + (a_2^2x_1^2 + a_1^2x_2^2) + (a_1 + a_2)x_1x_2\right)t^2 + \ldots \tag{5.56}$$

Interestingly enough this is just a formalization of the brute force method of determining the functional equation - by taking a previous set of walks, extending them by 1 step, removing the ones that fall outside of the quarter plane and finally labelling the ones that have interacted with the hyperplanes $x_i = 0$ in the process. We could attempt to write a closed form for the operator application but it becomes messy fairly fast, even for one dimension.

5.5 Kernel Form

We can write out the functional equations in a way that highlights the kernel K in the equation. This suggests that K is indeed special. We start out like

$$1 = Q - tLESQ = Q + LE(-tS + 1 - 1)Q \tag{5.57}$$
$$= Q + LEKQ - LEQ = (1 - L)Q + LEKQ \tag{5.58}$$

where we have $EQ = Q$ since Q only has non-negative powers of x_k. Next we can write

$$(1-L)Q + LEKQ = (1-L)Q + (LE-1)KQ + KQ \tag{5.59}$$

and finish off the calculation with

$$KQ + (1-L)Q - (1-LE)KQ = 1 \tag{5.60}$$

We can also start with the reduced form

$$\alpha = MQ - tESQ = MQ - EQ + EKQ = (M-1)Q + EKQ \tag{5.61}$$

here we can expand $E = E - 1 + 1$ to find

$$KQ = \alpha + (1-M)Q + (1-E)KQ = \alpha + (1-M)Q - t(1-E)SQ \tag{5.62}$$

where the last step follows from

$$(1-E)KQ = (1-E)(Q - tSQ) = -t(1-E)SQ \tag{5.63}$$

The operators $1-M$ and $1-E$ are purely extractions . The last kernel form shows more clearly where the K_I's come from in the functional equation above.

5.6 Walks without interactions

The Kernel forms above suggest an easy route for dealing with the models without interactions. Those models are equivalent to setting all the $a_k = 1$ which removes any marking or unmarking operators. The first kernel form reduces to the simple functional equation

$$E(KQ) = 1 \tag{5.64}$$

This suggests using the following substitution

$$Q = \frac{W}{x_{[d]}K} \tag{5.65}$$

so we have

$$x_{[d]} = x_{[d]}E(KQ) = x_{[d]}E\bar{x}_{[d]}W = \Gamma W \tag{5.66}$$

where $\Gamma = \Gamma_1 \dots \Gamma_d$ and

$$\Gamma_k = x_k E_k \bar{x}_k = x_k(1 - \bar{x}_k[\bar{x}_k])\bar{x}_k = 1 - [x_k]^0 \tag{5.67}$$

Therefore we have the functional equation for W as

Proposition 19. *Let $W = x_{[d]}KQ$ where Q is the generating function for walks without interactions in the positive orthant*

$$\Gamma W = \sum_{I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{O}} = x_{[d]} \tag{5.68}$$

For example in two dimensions we have

$$W(x_1, x_2) - W_1^0(x_2) - W_2^0(x_1) + W_{1,2}^{0,0} = x_1 x_2 \tag{5.69}$$

Thus we can write the solution to the problem as

$$Q = \frac{1}{x_{[d]}K} \left(x_{[d]} - \sum_{\emptyset \neq I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{O}} \right) \tag{5.70}$$

In two dimensions this becomes

$$Q(x_1, x_2) = \frac{x_1 x_2 + W_1^0(x_2) + W_2^0(x_1) - W_{1,2}^{0,0}}{x_1 x_2 (1 - tS)} \tag{5.71}$$

Of course each $W_I^{\mathbb{O}}$ depends on $Q_I^{\mathbb{O}}$ so we could have seen this relationship from the master functional equation for Q too. Interestingly enough this calculation points

to a differential equation. Clearly, W satisfies the differential equations

$$\partial_1 \partial_2 W = 1 \tag{5.72}$$

and in general

Proposition 20. *The function Q for walks without interactions is D-finite in the $x_1, \ldots, x_d$ variables since it satisfies the inhomogeneous PDE*

$$\partial_1 \ldots \partial_d W = \partial_1 \ldots \partial_d (x_1 \ldots x_d K Q) = 1 \tag{5.73}$$

We'll see a more general relationship in a later chapter.

Chapter 6

Operators

In this section we'll make our way towards the larger problem of higher-dimensional markled walks. The methods will be similar to those in the section on the one-dimensional problem but their complexity will rise substantially. Without the operator notation even the derivation of the functional equation is quite laborious. It's probably worth mentioning that we found the derivation the hard way first before we discovered we can use operators to simplify the process. What once took about a dozen pages now takes a just few. In this section we'll explore the properties of the operators that interest us, showing when the commute, if they have inverses, what their eigenvalues and eigenvectors are and so on.

6.1 Homogeneous Operators

6.1.1 Single 1D Operators

We start off with the observation that an extraction operator E is just a marking operator L with mark $a = 0$. In other words, we can combine the discussion for E and L by combining the operators into one definition.

Definition 21. *For integer n and a number a let*

$$\Gamma_n(a) = 1 + (a-1)x^n[x^n] = 1 + bx^n[x^n] \tag{6.1}$$

Lemma 22. *The operators $\Gamma_m(a_1)$ and $\Gamma_n(a_2)$ commute.*

Proof. Whether the operators mark the same monomial or not, since they act by multiplication by a_i it doesn't matter in which order we apply them. We can also see this computationally. If $m \neq n$

$$\Gamma_n(a_1)\Gamma_m(a_2) = 1 + b_1 x^n[x^n] + b_2 x^m[x^m] + b_1 b_2 x^n[x^n]x^m[x_m] \tag{6.2}$$
$$= 1 + b_1 x^n[x^n] + b_2 x^m[x^m] \tag{6.3}$$
$$= \Gamma_m(a_2)\Gamma_n(a_1) \tag{6.4}$$

The case for $m = n$ is analogous. □

Because of the commutativity we can write a product of such operators in any order we like. We'll usually write them in alphabetical order. Next, we'll find the eigenvalues and eigenfunctions for single operators.

We note further properties of $\Gamma_n(a)$. The first has to do with its eigenvalues and eigenfunctions. If $a = 1$ the operator acts as the identity so every function is an eigenfunction with eigenvalue 1. When $a \neq 1$ we see that the operator keeps all monomials the same except for the one it marks x^n which will be an eigenfunction with eigenvalue a. It's worth seeing this proved in a linear algebra style both to illustrate the method and to act as a consistency check.

Proposition 23. *The operator $\Gamma_n(a)$ has eigenvalues a and 1 with corresponding eigenfunctions x^n for a and x^k, $k \neq 1$ for 1.*

Proof. Suppose that for some function $u \neq 0$ and $a \neq 1$

$$\Gamma_n(a)u = \lambda u. \tag{6.5}$$

We can write $u = u_n x^n + v$ where v has no x^n term. Therefore we find

$$0 = \Gamma_n(a)u - \lambda u = a u_n x^n + v - \lambda u_n x^n - \lambda v = (a - \lambda)u_n x^n + (1 - \lambda)v \tag{6.6}$$

We have two cases

- If $v = 0$ then $u_n \neq 0$ and $\lambda = a$. The eigenfunction is x^n.
- If $v \neq 0$ then $\lambda = 1$ and $u_n = 0$. Here every x^k for $k \neq n$ is an eigenfunction.

□

The second interesting property has to do with taking powers of $\Gamma_n(a)$, including inverses.

Proposition 24. *We have*

$$\Gamma_n(a)^m = \Gamma_n(a^m) \tag{6.7}$$

which holds true for all integers m when $a \neq 0$ and all non-negative integers when $a = 0$. In particular $\Gamma_n(0)^m = \Gamma_n(0)$.

When $a = 0$ the operator has a non-trivial kernel generated by x^n. We can think of minimal polynomials too. For example $\Gamma_n(0)$ is non-trivial so its minimal polynomial is not X or $X-1$. However since it's idempotent, its minimal polynomial must divide $X(X-1)$ and since it's neither of the factors it must be the minimal polynomial. On the other hand $\Gamma_n(a)$ for $a \neq 0$ does not have a minimal polynomial because the powers of a are independent from each other.

6.1.2 Compounded 1D Operators

Naturally we are led to consider compounded operators and their properties. Let $\Gamma = \Gamma_{n_1}(a_1)\dots\Gamma_{n_m}(a_m)$ where the powers n_i are distinct; otherwise treat any repetitions using commutativity and the identity $\Gamma_n(a)^m = \Gamma_n(a^m)$. Using induction with the identity above gives

$$\Gamma = 1 + b_1 x^{n_1}[x^{n_1}] + \cdots + b_m x^{n_m}[x^m] \tag{6.8}$$

Letting $u = u_{n_1}x^{n_1} + \cdots + u_{n_m}x^{n_m} + v$ in the same way as the single operator case and going through the calculation shows that the eigenvalues of Γ are

- $\lambda = a_i$ for $1 \leq i \leq m$ with eigenfunction x^{n_i}.
- $\lambda = 1$ with eigenfunctions x^m where $m \neq n_i$ for any i.

When one or more of the a_i's is zero we have a non-trivial kernel

$$\ker(\Gamma) = \langle x^m | m = n_i, a_i = 0 \rangle \tag{6.9}$$

6.1.3 Multidimensional Operators

Multidimensional operators are ones composed of one-dimensional operators (whether single or compounded) and they can also be considered compounded. For this section we'll use superscript notation

$$\Gamma_n^{(k)}(a) = 1 + bx_k^n[x_k^n] \tag{6.10}$$

Operators working on different dimensions commute with each other

$$\Gamma_n^{(i)}(a_1)\Gamma_m^{(j)}(a_2) = 1 + b_1 x_i^n[x_i^n] + a_2 x_j^m[x_j^m] + a_1 a_2 x_i^n x_j^m[x_i^n x_j^m] \tag{6.11}$$

$$= \Gamma_m^{(j)}(a_2)\Gamma_n^{(i)}(a_1). \tag{6.12}$$

We can write $\Gamma^{(k)} = \Gamma_{n_1}(a_{k,n_1}) \dots \Gamma_{n_r}(a_{k,n_r})$ as before. This allows us to write a multidimensional operator in terms of its constituent pieces in whichever order we prefer. We are of course interested in finding eigenvalues and eigenfunctions for such operators. We could either find those with the same method as before or we could find it via induction.

Proposition 25. *All monomials $x_{[d]}^P$ are eigenfunctions of $\Gamma = \Gamma^{(1)} \dots \Gamma^{(d)}$. Their eigenvalues are determined by the action of $\Gamma^{(k)}$ on $x_k^{P_k}$. That is, the eigenvalues are products of the eigenvalues produced by the single-dimensional operators.*

This helps us characterize the kernel of Γ

Corollary 26. *The kernel of Γ is generated by those monomials $x_{[d]}^P$ for which $\Gamma^{(k)} x_k^{P_k} = 0$ for some k.*

6.2 Inhomogeneous Operators

We'll call an operator inhomogeneous if it consists of a homogeneous operator Γ and a laurent polynomial (possible infinite) S and and acts in one of the following ways

$$\Gamma S(u) = \Gamma(Su) \quad \text{or} \quad S\Gamma u = S \cdot \Gamma u \tag{6.13}$$

This kind of operator is important because it features quite a bit in the funcitonal equations of random walk models. We seek to find the eigenpairs for inhomogeneous operators. Our first task is to show that these operators are equivalent.

Proposition 27. *The two operators have the same eigenvalues with eigenfunctions differing by a multiple of S. That is* (λ, u) *is an eigenpair for* ΓS *if and only of* (λ, Su) *is an eigenpair for* $S\Gamma$*. In particular, if* $\lambda = 0$ *we have*

$$\ker(\Gamma S) = \overline{S}\ker(\Gamma) \tag{6.14}$$

Proof. Note that

$$\Gamma S u = \lambda u \quad \Longleftrightarrow \quad S\Gamma(Su) = \lambda(Su) \tag{6.15}$$

Therefore the operators have the same eigenvalues and they have the eigenfunctions u and Su respectively. Now if we pick $\lambda = 0$ to find the kernels of the operators we find that $u \in \ker(\Gamma S)$ if and only if $Su \in \ker(S\Gamma) = \ker(\Gamma)$ which completes the proof. □

6.2.1 Single 1D Operators

Let G and u be given as

$$S = \sum_k S_k x^k, \quad u = \sum_k u_k x^k \tag{6.16}$$

where u is an eigenfunction of ΓS. We have

$$0 = (\Gamma S - \lambda I)u \tag{6.17}$$

$$= \sum_{k \neq n} x^k \left(\sum_i S_i u_{k-i} - \lambda u_i \right) + x^n \left(a \sum_i S_i u_{n-i} - \lambda u_n \right) \tag{6.18}$$

which leads to the system of equations

$$\lambda u_n = a \sum_i S_i u_{n-i} \tag{6.19}$$

$$\lambda u_k = \sum_i S_i u_{k-i}, \quad \text{if } k \neq n \tag{6.20}$$

The analysis is non-trivial even for a simple function S like $S = x$. We have

$$\lambda u_n = a u_{n-1} \tag{6.21}$$

$$\lambda u_k = u_{k-1}, \quad \text{if } k \neq n \tag{6.22}$$

We have a doubly-infinite recursion so we can pick an arbitrary initial condition. If $u_n = 0$ the rest of the terms will be zero as well so we must have $u_n \neq 0$. Set $u_n = 1$. Then via induction we can find

$$u_{n+k} = 1/\lambda^k, \quad \text{for } k \geq 0 \tag{6.23}$$

$$u_{n-k} = \lambda^k / a^k, \quad \text{for } k \geq 1 \tag{6.24}$$

Computing $u(x)$ formally leads to

$$u(x) = \sum_{k=1}^{\infty} u_{n-k} x^{n-k} + \sum_{k=0}^{\infty} u_{n+k} x^{n+k} \tag{6.25}$$

$$= x^n \sum_{k=1}^{\infty} \left(\frac{\lambda}{ax}\right)^k + x^n \sum_{k=0}^{\infty} \left(\frac{x}{\lambda}\right)^k \tag{6.26}$$

$$= \frac{x^{n-1} x}{ax - \lambda} - \frac{x^n \lambda}{x - \lambda} \tag{6.27}$$

$$= \lambda x^n \left(\frac{1}{\lambda - x} - \frac{1}{\lambda - ax}\right) \tag{6.28}$$

$$= -\frac{\lambda b x^{n+1}}{(\lambda - x)(\lambda - ax)} \tag{6.29}$$

We could also try using the other definition. Let $u = u_n x^n + v$ where u is now

an eigenfunction of $S\Gamma$. Let $S = S_0 + S'$. Then we have

$$S\Gamma u = (S_0 + S')(au_n x^n + v) = \lambda(u_n x^n + v) \tag{6.30}$$

$$0 = (au_n S_0 - \lambda u_n + \sum_{k \neq n} S_k u_{n-k})x^n + au_n S' x^n + S' v - \lambda v \tag{6.31}$$

We find that for example

$$(\lambda - aS_0)u_n = \sum_{k \neq n} S_k u_{n-k} \tag{6.32}$$

which doesn't seem to help much.

6.3 Random Walks Operator

In this section we explore the operator $\mathscr{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some random walk operator Γ. We haven't made it rigorous yet but $\mathscr{L}$ is invertible with inverse

$$\mathscr{L}^{-1} = \sum_{k=0}^{\infty} t^k (\Gamma S)^k \tag{6.33}$$

where $(\Gamma S)^k$ is an iteration of the inhomogeneous operator with $(\Gamma S)^0 = 1$. However this inverse is not easy to use as the iteration become quite complicated due to the nature of the random walks. Let (u, λ) be an eigenpair for $\mathscr{L}$.Then we calculate

$$(\mathscr{L} - \lambda I)u = u - \lambda u - t\Gamma S u \tag{6.34}$$

$$t\Gamma S u = (1 - \lambda)u = \mu u \tag{6.35}$$

If $\lambda = 1$ we seek to find the kernel of ΓS which we did in the previous section. The eigenfunctions of $\mathscr{L}$ for eigenvalue $\lambda = 1$ are the functions $u \in \overline{S} \ker \Gamma$.

Next let u be given by a finite order Laurent series

$$u = u_n t^n + u_{n+1} t^{n+1} + \dots \tag{6.36}$$

where $u_n \neq 0$. Then we can re-write the functional equation as

$$t^{-n}(1-t\Gamma S)u = \lambda t^{-n}u. \tag{6.37}$$

Letting $t \to 0$ in this equation yields

$$(\lambda-1)u_n = -\lim_{t\to 0} t^{1-n}\Gamma S u = -\Gamma S \lim_{t\to 0}(t^{1-n}u) = 0. \tag{6.38}$$

Since $u_n \neq 0$ this implies that $\lambda = 1$ and again we see that $u \in \overline{S}\ker\Gamma$. Since S is a Laurent polynomial, the resulting functions will indeed be finite order Laurent series. For the remainder of the analysis we assume that $\lambda \neq 1$. This excludes finite order Laurent series so u is necessarily given by an infinite order Laurent series.

6.4 Interaction with ∂_t

Here we'll investigate ∂_t interacts with $\mathscr{L}$.

$$\partial_t \mathscr{L} = \partial_t - \partial_t(t\Gamma S) = \partial_t - \Gamma S - t\Gamma S\partial_t = \mathscr{L}\partial_t - \Gamma S \tag{6.39}$$

$$\partial_t^2 \mathscr{L} = \mathscr{L}\partial_t^2 - 2\partial_t\Gamma S \tag{6.40}$$

$$\partial_t^n \mathscr{L} = \mathscr{L}\partial_t^n - n\partial_t^{n-1}\Gamma S \tag{6.41}$$

for integer n. Another way to write these is with bracket notation

$$[\mathscr{L}, \partial_t^n] = \mathscr{L}\partial_t^n - \partial_t^n \mathscr{L} = n\partial_t^{n-1}\Gamma S \tag{6.42}$$

6.5 Interaction with ∂_i

Let's take the operator $\Gamma_n(a) = 1 + bx^n[x^n]$ again. Let $\partial_k = \partial/\partial_k$ and in this case just ∂. We have

$$\partial\Gamma_n(a) = \partial + b\partial(x^n[x^n]) = \partial + bnx^{n-1}[x^n] \tag{6.43}$$

Now if $n \geq 0$ we have

$$\partial^{n+1}\Gamma_n(a) = \partial^{n+1} + b\partial^{n+1}(x^n)[x^n] = \partial^{n+1} \tag{6.44}$$

If $n \leq 0$ taking derivatives will yield ever greater powers of $\bar{x}$ so we want to pre-multiply by a power of x first.

$$\partial x^{-n}\Gamma_n(a) = \partial x^{-n} + b\partial[x^n] = \partial x^{-n} = -nx^{-n-1} + x^{-n}\partial \tag{6.45}$$

If we have more terms in an operator then we need to multiply by a power of x to clear out any negative powers of x. Then we take the lowest order derivative in order to clear out all the extraction terms. Suppose that

$$\Gamma = 1 + \sum_{i=0}^{m} a_i x^i[x^i] - \sum_{i=1}^{n} b_i \bar{x}^i[\bar{x}^i] \tag{6.46}$$

where m and n are non-negative. We need to pre-multiply by x^n to clear out all the negative powers of x. Then we have

$$x^n\Gamma = x^n + \sum_{i=0}^{m} a_i x^{i+n}[x^{i+n}] - \sum_{i=1}^{n} b_i x^{n-i}[x^{n-i}] \tag{6.47}$$

This operator is cleared by ∂^{m+n+1}

Chapter 7

Discussion on Random Walk Model Generalizations

7.1 Introduction

The operator notation makes it easy to handle more complicated random walk problems and in this section we'll attempt ot push the notation to its limits. Let's suppose we would like to enumerate random walks constrained to be within a region $\mathscr{D}$ in space. We would also like to mark any interactions with certain locations $\mathscr{D}_0$ within $\mathscr{D}$. To do this we can start as before by extending the walks enumerated by Q by one step. We do this by calculating the product SQ. Some walks venture outside of the region $\mathscr{D}$ so we remove them with an operator E. Then we count all the interactions by marking walks ending up on $\mathscr{D}_0$ using an operator operator L. Since we have an extra step now, we also need to multiply by t. We obtain the expression $tLESQ$. As before we need to find out how this expression relates to Q. The only walks we've omited are those of length 0. That is if $[t^0]Q = q$ we have

$$(1 - tLES)Q = q \tag{7.1}$$

We can also write this as

$$(L^{-1} - tES)Q = L^{-1}q = \alpha \tag{7.2}$$

We have obtained the functional equation for the generating function that enumerates random walks

- beginning in locations enumerated by q
- evolving with a step Laurent polynomial S
- constrained to evolve in a region $\mathscr{D}$
- interacting with region $\mathscr{D}_0 \subseteq \mathscr{D}$

The second form of the functional equation should have less terms than the first one by virtue of separating L and E from one another. Using extraction operators we can constrain random walks to lie within a finite region, semi-infinite strip, a semi-infinite sector or any other discrete shape (with diminishing returns for more complicated shapes of course). Furthermore using marking operators we can easily track the behavior of random walks as they cross or interact with boundaries or interact with regions within. For example we may consider walks interacting with the boundary of a box within a region in space. We can also take on the problem from another perspective. Instead of removing walks that stray outside of the boundary of $\mathscr{D}$ we can just mark them for later deletion. This allows us to consider a more holistic problem where we combine the operators L and E into one called Γ. That is in the case where we mark some walks for deletion with parameter a we obtain a functional equation

$$(1 - t\Gamma S)\tilde{Q} = q \tag{7.3}$$

where $\tilde{Q} = Q + O(a)$ and Q is the generating function in the previous functional equations. We just need to extract $[a^0]$ from $\tilde{Q}$ to obtain Q. Since the two problems are equivalent we can use the latter to combine the discussion of the marking operators. Before we proceed we should mention that a previous discussion on the kernel form applies here as well since it was mostly a symbolic calculation. We have

$$q = (1 - tLES)Q = Q - t(LE - 1 + 1)SQ = (1 - tS)Q - t(LE - 1)(SQ) \tag{7.4}$$

so

$$KQ = q - t(1-LE)SQ. \tag{7.5}$$

The second kernel form is obtained via

$$\alpha = (L^{-1}-1)Q + Q - tSQ - t(E-1)SQ = KQ - (1-L^{-1})Q + t(1-E)SQ \tag{7.6}$$

which leads to

$$KQ = \alpha + (1-L^{-1})Q - t(1-E)SQ \tag{7.7}$$

If we want to emphasize the relationship with KQ itself we have

$$KQ = L^{-1}(q) + (1-L^{-1})Q + (1-E)KQ \tag{7.8}$$

which follows from $(1-E)Q = Q - Q = 0$. If we decide not to mark any of the walks we have the simpler Kernel form

$$(1-tES)Q = Q - EQ + E(KQ) = E(KQ) \tag{7.9}$$

so we have the functional equation

$$E(KQ) = q \tag{7.10}$$

which as before suggests writing $Q = \tilde{W}/K$ which yields

$$E(\tilde{W}) = q \tag{7.11}$$

as a functional equation defining the unmarked generating function. The beauty of all these equations is that they are all relative to an arbitrary domain $\mathscr{D}$ with interaction domain $\mathscr{D}_0$.

7.2 Marking Operators

We'll take care of regional constraints and interactions in one discussion. Suppose that we wish to mark points $p \in P \subseteq \mathbb{Z}^d$ with labels a_p via an operator Γ_P. It can be written as

$$\Gamma_P = 1 + \sum_{p \in P} (a_p - 1) x^p_{[d]} [x^p_{[d]}] \tag{7.12}$$

The form $(a_p - 1)$ follows from the fact that we have the original term and we need to remove it before marking with a_p. In the case where we mark points for deletion $P = \mathscr{D}^c$ and $a_p = a$. The good thing is that in a lot of cases we don't need all the terms in the operator and in some cases we even find the operator has a portion that factors. We just need to ensure that we mark all the walks that touch the boundaries or go within the region marked for deletion. For example we can take the walks in one dimensions on the non-negative axis with steps of size at most n in the negative direction. Then we can demonstrate the truncation with

$$\Gamma = 1 + (a-1) \sum_{k=1}^{\infty} \bar{x}^k [\bar{x}^k] = 1 + (a-1) \sum_{k=1}^{n} \bar{x}^k [\bar{x}^k] \tag{7.13}$$

We only need to consider how far a walk can step outside of the region $\mathscr{D}$ via the step set in S. In two dimensions we have

$$\Gamma = 1 + (a-1) \sum_{m \text{ or } n<0}^{\infty} \bar{x}_1^m \bar{x}_2^n [\bar{x}_1^m \bar{x}_2^n] \tag{7.14}$$

Without a this reduces to

$$\Gamma = 1 - \bar{x}_1 [\bar{x}_1] - \bar{x}_2 [\bar{x}_2] + \bar{x}_1 \bar{x}_2 [\bar{x}_1 \bar{x}_2] = (1 - x[x]_1)(1 - x[x]_2) \tag{7.15}$$

in which case we also see that the operator can factor into two or more operators sometimes. When the boundary is complicated we may not find an a finite form for Γ let alone see any factoring. However, there are some nice special cases where we can say a lot about the operator Γ. We can build the theory. We have only finitely many steps (and sizes of steps) in S. We say Γ is finite if it can be represented by finitely many coefficient extraction terms. Suppose that $\mathscr{D}$ is a finite box in

$\mathbb{Z}^d$. Then Γ is finite. To see this suppose that B has x_k coordinates in the interval $[a_k, b_k]$ and that stepping outside from B from within B via step polynomial S leads to x_k coordinates within the interval $[m_k, n_k]$. Then Γ can be written finitely with extraction part

$$\Gamma_{B^c} = \sum_{k=1}^{d} \left(\sum_{i=m_k}^{a_k-1} x_k^i [x_k^i] + \sum_{i=b_k+1}^{n_k} x_k^i [x_k^i] \right) \tag{7.16}$$

The discussion continues for semi-infinite and infinite boxes with finite marked regions within. The analysis is the same as above except that now one or more of a_k, b_k may be infinite, in which case we actually lose parts of Γ_{B^c} and the operator is still finite. We should see if we can do regions whose complements are boxes and then build up to finite regions and intersections of finite regions with boxes.

Definition 28. *A coefficient extraction operator Γ in the variables $x_1, \ldots, x_d$ is of finite type if it can be written in a form with only finitely man terms like*

$$\Gamma = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J} x^J [x_I^J] \tag{7.17}$$

where each $|J| < \infty$. Then Γ is of infinite type if it's not of finite type. We'll just say Γ is finite or infinite.

Definition 29. *A random walk model with finitely many steps is of finite type if its extraction and marking operator Γ is of finite type and its starting condition q is of finite degree. If not, the model is of infinite type.*

Not surprisingly all the models so far has been of finite type. In the next section we'll see the model of self-avoiding walks is most likely of infinite type.

Theorem 30. *The generating function Q for a random walk model of finite type is D-finite in the variables $x_1, \ldots, x_d$.*

Proof. We'll use the kernel form for the functional equation

$$KQ + t(1-\Gamma)SQ = q. \tag{7.18}$$

Here $\Gamma' = 1 - \Gamma$ is purely a coefficient extractor and is of finite type.We'll prove this by finding the differential equation explicitly. Since the model is of finite type we can bound the degree of x_k that we see on either side of the functional equation. In one scheme we let A_k be the largest power of x_k that we need to clear a negative power of x_k in Γ' and in q. Then we let B_k be the largest power of x_k that we see on either side. Once we multiply by $x_k^{A_k}$ we see that the degree of x_k in the equation is $A_k + B_k$. We take the functional equation and multiply both sides by $x_{[d]}^A$

$$x_{[d]}^A(KQ + t\Gamma' SQ) = x_{[d]}^A q \tag{7.19}$$

and we obtain a polynomial equation in the variables $x_1, \ldots, x_d$. We take the derivative operator

$$D = \partial_{[d]}^{A+B+\mathbb{I}} = \partial_1^{A_1+B_1+1} \ldots \partial_d^{A_d+B_d+1} \tag{7.20}$$

and apply it to both sides of the equation to get

$$D(x_{[d]}^A(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x_{[d]}^A q) = 0 \tag{7.21}$$

which follows since we're taking more derivatives than the degrees of the monomials which clears all of them. The other scheme lets us obtain a potentially lower degree differential equation at the cost of it beind inhomogeneous. Let A_k be the power of x_k that we need to clear denominators on the LHS of the functional equation and B_k be the largest power of x_k on that side. Then we have D as before but this time it may have lower A and B values since we are not taking q into account. We have

$$D(x_{[d]}^A(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x_{[d]}^A q) \tag{7.22}$$

□

For example let the powers of x_k be $\{-3,-1,4\}$ and $\{-2,0,6\}$ on the LHS and RHS respectively. We see that $A_k = 3$ and $B_k = 6$. Then we're taking the $A_k + B_k + 1 = 10$-th derivative of both sides of the functional equation. In The second scheme we take $A_k = 3$ still but $B_k = 4$ and we are taking the 8-th derivative

w.r.t. x_k this time. Even in the quarter-plane we have models that have a generating function that is not D-finite (with respect to the t variable) but it turns out that every model so far considered is D-finite in the $x_1, \ldots, x_d$ variables.

In order to solve the differential equation we need to investigate the kernel of the differential operator.

Corollary 31. *The generating function Q of a finite random model is given by*

$$x_{[d]}^{A} KQ(x_1, \ldots, x_d; t) = \sum_{i=1}^{d} \sum_{j=0}^{B_i} Aij(x_1, \ldots, x_{i-1}, x_i, \ldots, x_d, t) x_i^j \tag{7.23}$$

where A_{ij} are functions that need to be determined.

We could have also obtained this kind of solution from going through the calculations in the functional equation and extracted coefficients. However, this method is more efficient. If we had used the second scheme in the proof we would have ended up with a slightly different form but maybe it would have had a slightly lower degree. It would involve an indefinite integral of q and elements in the kernel of the differential operator.

7.3 Self-avoiding walks

We can handle self-avoiding random walks with this theory. If we pick a marking operator Γ that marks every location on the plane this will allow us to track the random walks' history. All we'll need to do in the end is remove those walks which have stepped at some location more than once. However we'll do this problem in a bit more detail as it serves as an example of the theory. To use the generating function iteration properly we need a sort of limiting argument. For the first n steps we "only" require an operator covering the square region of $(2n+1)^2$ lattice points. The overall operator is defined through a limiting process of applying the full operator (or truncated version) on a finite Laurent polynomial. Let

$$\Gamma_{i,j} = 1 + b_{i,j} x_1^i x_2^j [x_1^i x_2^j] \tag{7.24}$$

where as usual $b_{i,j} = a_{i,j} - 1$. This operator takes a generating function and marks the monomial $x_1^i x_2^j$. We'll be using a lot of these together so we let

$$\Gamma_n = \prod_{-n \le i,j \le n} \Gamma_{i,j} \tag{7.25}$$

To find a functional equation for SAWs on the plane we perform an iteration. Let Q be the generating function for walks with history, that is they have a mark for each coordinate they pass through and let $Q^{\le n}$ be the truncated version of this generating function. It's given by

$$Q^{\le n} = \sum_{k=0}^{n} Q_k t^k \tag{7.26}$$

where the initial condition is $Q_0 = a_{0,0}$. Starting from Q_n we can find Q_{n+1} by iterating with the step set S, marking with Γ_{n+1} as follows

$$Q_{n+1} = \Gamma_{n+1} S Q_n \tag{7.27}$$

Using a computer we can easily iterate this to get more and more steps. We can also get $Q^{\le n+1}$ from $Q^{\le n}$ since

$$t\Gamma_{n+1} S Q^{\le n} = \sum_{k=0}^{n} \Gamma_{n+1} S Q_k t^{k+1} \tag{7.28}$$

$$= \sum_{k=0}^{n} \Gamma_{k+1} S Q_k t^{k+1} \tag{7.29}$$

$$= \sum_{k=0}^{n} Q_{k+1} t^{k+1} \tag{7.30}$$

$$= Q^{\le n+1} - Q_0 \tag{7.31}$$

or in other words

$$Q^{\le n+1} = a_{0,0} + t\Gamma_{n+1} S Q^{\le n} \tag{7.32}$$

In the interest of finding a functional equation we would like to take the limiit of this as $n \to \infty$. For this we define Γ to be the operator marking all monomials $x_1^i x_2^j$

and acting in the following way

$$\Gamma x^i y^j = \Gamma_{\max(|i|,|j|)} x^i y^j \tag{7.33}$$

and extended by linearity. The limiting process is

$$Q = a_{0,0} + t\Gamma S Q \tag{7.34}$$

We finally have the functional equation for random walks with full history

$$(1 - t\Gamma S)Q = a_{0,0}. \tag{7.35}$$

From this we can obtain the generating function W for the SAWs by removing all walks with power of mark $a_{i,j}$ greater than 1. For this we need a removal operator. Let

$$R_{i,j} = a^0_{i,j}[a^0_{i,j}] + a_{i,j}[a_{i,j}] \tag{7.36}$$

$$R_n = \prod_{-n \le i,j \le n} R_{i,j} \tag{7.37}$$

$$R = \lim_{n \to \infty} R_n \tag{7.38}$$

where the limit is understood as in the example of Γ above. The solution for Q can also be restated as

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (\Gamma_k S)^k (a_{0,0}) \tag{7.39}$$

We can either redo the analysis applying R at every iteration or we can apply R to Q and get

$$W = RQ = \sum_{k=0}^{\infty} t^k R(\Gamma S)^k (a_{0,0}) = \sum_{k=0}^{\infty} t^k (R\Gamma S)^k (a_{0,0}) \tag{7.40}$$

We can write this as a functional equation

$$\mathscr{L} W = (1 - t\Sigma)W = (1 - tR\Gamma S)W = a_{0,0} \tag{7.41}$$

7.3.1 Eigenpairs

We can try finding eigenvalues and eigenfunctions of $\mathscr{L}$. Let $\mathscr{L}u = \lambda u$. Writing it out leaves us with

$$t\Sigma u = (1-\lambda)u \tag{7.42}$$

So if $\lambda = 1$, $u \in \ker(\Sigma)$. That is $R\Gamma Su = 0$. In other words ΓSu has only terms that involve powers greater than 1 of $a_{i,j}$ for various i, j. Suppose that

$$\Gamma Su = a_{i,j}^2 f(x_1, x_2) \tag{7.43}$$

Then

$$u = \frac{a_{i,j}^2}{S}\Gamma^{-1}(f) \tag{7.44}$$

So we here we can let

$$f = a_{k,l}^m x_1^r x_2^s t^n \tag{7.45}$$

and we find that

$$u = \frac{a_{i,j}^2 a_{k,l}^m}{a_{r,s}S} x_1^r x_2^s t^n \tag{7.46}$$

Thus the kernel of Σ is generated by these functions for all coefficients $i, j, k, l, r, s, n \in \mathbb{Z}$ and $m \geq 0$. This may perhaps prove useful later on. Now, if we look at eigenvalues $\lambda \neq 1$ we have to solve the equation

$$t\Sigma u = (1-\lambda)u \tag{7.47}$$

Because of the t, if u is a finite order laurent series in t with first non-zero coeficient u_n. We find that since Σ does not affect the t monomials

$$\Sigma u_n = \mu u_{n-1} = 0 \tag{7.48}$$

and this forces all u_k coefficients to be zero. Therefore Σ has no other eigenvalues except 1 among the finite order power series in t.

7.3.2 Generalizations

The discussion lends itself nicely to generalization where now we have d variables and operators $\Gamma_{i_1,\ldots,l_d}$, $R_{i_1,\ldots,i_d}$, etc defined analogously. The eigenpair discussion works out in the same way with eigenfunctions for $\lambda = 1$

$$u = \frac{a^2_{i_1,\ldots,i_d} a^m_{j_1,\ldots,j_d}}{a_{k_1,\ldots,k_d} S} x_1^{r_1} \ldots x_d^{r_d} t^n \tag{7.49}$$

We can also constrain the SAWs on a part of the plane with an extraction operator E and mark special locations with an operator L. The functional equation is

$$(1 - tLE\Sigma)W = a_{0,\ldots,0} \tag{7.50}$$

for arbitrary number of dimensions, with region constraints and marking of special locations.

7.4 Coefficient Extraction Equations

The work so far is strongly hinting towards a theory of ordinary and partial coefficient extraction equations - OCEs and PCEs. Here we'll work out some of the theory. We start with a partial coefficient extraction operator of finite type

Definition 32. *$\mathscr{L}$ is a coefficient operator if*

$$\mathscr{L} = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J} [x_I^J] = K + \mathscr{E} \tag{7.51}$$

where K is the kernel of the operator, $\mathscr{E}$ is the pure extraction part

$$\mathscr{E} = \sum_{\substack{I \subseteq [d], I \neq \emptyset \\ J \subseteq \mathbb{Z}^d}} a_{I,J} [x_I^J] \tag{7.52}$$

and all the coefficients are Laurent polynomials in the x variables and perhaps

some other variables. The operator is ordinary if $d = 1$ and partial if $d \geq 2$. The operator is either of monomial type where the coefficients are monomials or of polynomial type if one or more coefficients is a Laurent polynomial in $x_1, \ldots, x_d$.

Definition 33. *An ordinary/partial coefficient equation (OCE/PCE) is one of the form $\mathscr{L} Q = q$ for some functions Q and q where $\mathscr{L}$ is an ordinary/partial coefficient operator. The equation is homogeneous if $q = 0$ and inhomogeneous if $q \neq 0$.*

The question is whether we can solve these kinds of equations in general and what their solutions may look like. In the previous section we found that if $\mathscr{L}$ is finite then the function Q is D-finite in the x variables while it may not necessarily be D-finite in the t variable or any other variable (parameter) that appears in the coefficients.

Example 34. The operator Γ in this section is a partial coefficient operator with monomial coefficients.

Example 35. We can transform the functional equation for a finite random walk model into a PCE. To do so we note that K is the same in both forms. The part requiring some work is the one that has to do with the term $(1-\Gamma)SQ$ which is a pure extraction of SQ. We can expand S into its monomials. Then $(1-\Gamma)S$ fully expands into monomials of the form

$$a_{I,J} x_I^J [x_I^J] x_A^B \tag{7.53}$$

we can pull terms inside $[x_I^J]$ from the right. In the end the S is absorbed within the operator and istead of having monomial coefficients it now has polynomial coefficients. It may be useful to write an operator $\mathscr{L}$ in the form

$$\mathscr{L} = K + \Gamma' S \tag{7.54}$$

for some monomial operator Γ'.

Example 36. Here is an example that may hint at the flavour of the OCE/PCE

problems. Suppose we start with the fibonacci recurrence relation

$$F_{n+2} = F_{n+1} + F_n \tag{7.55}$$

with some initial conditions for f_0 and f_1. We can perform the standard generating function solution approach via summing all equations. Let

$$F(x) = \sum_{n=0}^{\infty} F_n x^n \tag{7.56}$$

be the generating function for the fibonacci-like numbers. Note that $[x^n]F(x) = F_n$. We have

$$F_{n+2} = F_{n+1} + F_n \tag{7.57}$$

$$F_{n+2}x^{n+2} = F_{n+1}x^{n+2} + F_n x^{n+2} \tag{7.58}$$

$$\sum_{n=0}^{\infty} F_{n+2}x^{n+2} = x\sum_{n=0}^{\infty} F_{n+1}x^{n+1} + x^2\sum_{n=0}^{\infty} F_n x^n \tag{7.59}$$

$$F - F_1 x - F_0 = x(F - F_0) + x^2 F \tag{7.60}$$

Pulling like terms together and putting everything on one side gives

$$(1 - x - x^2)F - xF_1 + (x-1)F_0 = 0 \tag{7.61}$$

This is a homogeneous OCE with ordinary coefficient operator $\mathscr{L}$ (Fibonacci operator?) with kernel $K = 1 - x - x^2$ and polynomial coefficients

$$\mathscr{L} = (1 - x - x^2) + (x-1)[x^0] - x[x] = K + \mathscr{E} \tag{7.62}$$

The typical solution to the problem specifies initial conditions F_0 and F_1 and solves the problem as

$$F(x) = \frac{(x-1)F_0 - xF_1}{1 - x - x^2} = \frac{\mathscr{E}(F)}{K} \tag{7.63}$$

That is, it relies on the initial condition for $\mathscr{E}$ in order to solve the equation. We also see that if we don't specify the initial conditions we don't get a unique solution. In

the random walks models we have unique solutions because of the iterative nature of the problems. With these points in mind we may try the vanishing kernel method by supplying a root of K namely $x = \sigma$. These roots are

$$\sigma_1 = -\varphi, \sigma_2 = \overline{\varphi} \tag{7.64}$$

where φ is the golden ratio. Then plugging either one of these into the OCE leads to

$$(\sigma - 1)F_0 = \sigma F_1 \tag{7.65}$$

$$F_1 = (1 - \overline{\sigma}_1)F_0 = \varphi F_0 \tag{7.66}$$

$$\text{or } F_1 = (1 - \overline{\sigma}_2)F_0 = -\overline{\varphi} F_0 \tag{7.67}$$

That is, if we want the vanishing kernel method to work we must have that F_1 is related to F_0 directly via the golden ratio. By induction we can show that $F_n = \varphi^n F_0$ in the first case and $F_n = (-\overline{\varphi})^n F_0$ in the second case.

What about the persistent kernel method? We want to apply a transformation $y = y(x)$ to K that leaves it the same. That is

$$K(y) = 1 - y - y^2 = K(x) = 1 - x - x^2 \tag{7.68}$$

Collecting terms on one side yields

$$(y - x)(1 + x + y) = 0 \tag{7.69}$$

That is, the group G that preserves the kernel is composed of the identity transformation and the the transformation $g(x) = -x - 1$. We want a $F(-1 - x)$ to converge which may not necessarily be true. Instead we multiply the whole functional equation by $\bar{x}$ to find

$$\bar{x}\mathscr{L} = (\bar{x} - 1 - x) + (1 - \bar{x})[x^0] - [x^1] \tag{7.70}$$

This new kernel K is nicer. We have

$$K = \bar{x} - 1 - x = \bar{y} - 1 - y \tag{7.71}$$

rearranging this equality gives

$$(y-x)(1+\overline{xy}) = 0 \tag{7.72}$$

and thus the group G is generated by the identity transformation and $g(x) = -\bar{x}$ which is of order 2, so $|G| = 2$. We have the two equation

$$KF + (1-\bar{x})F_0 - F_1 = 0 \tag{7.73}$$

$$KgF + (1+x)F_0 - F_1 = 0 \tag{7.74}$$

Furthermore emulating the orbit method we have

$$(1-g)xKF = K(xF + \bar{x}gF) = (x+\bar{x})(F_1 - F_0) \tag{7.75}$$

If $F_1 \neq F_0$ we can write

$$xF + \bar{x}gF = \frac{(x+\bar{x})}{K}(F_1 - F_0) \tag{7.76}$$

$$= \frac{1+x^2}{1-x-x^2}(F_1 - F_0) \tag{7.77}$$

$$= (1 + x + 3x^2 + O(x^3))(F_1 - F_0) \tag{7.78}$$

Extracting the $[x]$ coefficient yields

$$F_0 + [x^2]F(-\bar{x}) = F_0 = F_1 - F_0 \tag{7.79}$$

so $F_1 = 2F_0$ and we find

$$xF + \bar{x}F(-\bar{x}) = F_0 \frac{1+x^2}{1-x-x^2} \tag{7.80}$$

Taking the positive part of this yields

$$xF = F_0 \frac{1+x^2}{1-x-x^2} \tag{7.81}$$

I think there is something wrong with the example but that's just me. We can demonstrate the D-finiteness by taking the second derivative of xKF

$$\partial_x^2((1-x-x^2)F - xF_1 + (x-1)F_0) = \partial_x^2(xKF) = 0 \tag{7.82}$$

7.5 Connection with Fredholm Theory

Suppose that we are given the equation in the form

$$KQ + t\Gamma' SQ = q \tag{7.83}$$

where

$$\Gamma' = \sum_{I,J} a_{I,J} x_I^J [x_I^J] \tag{7.84}$$

where we only have finite-sized J's and $\emptyset \neq I \subseteq [d]$. From complex analysis we know that for a function $f(x)$ given by a power series and for x within the region of convergence

$$[x^n]f(x) = f_n = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z^{n+1}}dz \tag{7.85}$$

and

$$f(x) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-x}dz \tag{7.86}$$

where in the first case γ winds CCW around the origin within the region of convergence and in the second case γ winds around x within the region of convergence. The idea is to combine these two expressions for many dimensions (at least for-

mally) to represent the operator $\mathscr{L}$ in integral form. We have for $|I| = n$

$$[x_I^J]f(x_1,\ldots,x_d) = \frac{1}{(2\pi i)^n}\int\cdots\int_\gamma \frac{f}{z_I^{J+1}}dz_I \tag{7.87}$$

where f has $x_i = z_i$ for $i \in I$ and the rest are still x's. If we use this representation we'll have different numbers of iterations of the integral for different coefficient. We might as well go all-out and write

$$[x_I^J]f(x_1,\ldots,x_d) = \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \frac{f}{z_I^{J+1}(z-x)_{I^c}}dz_{[d]} \tag{7.88}$$

where now $x_i = z_i$ in f and

$$(z-x)_P = \prod_{p\in P}(z_p - x_p) \tag{7.89}$$

This allows us to write The operator like this

$$(K+t\Gamma' S)Q = \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \left(\frac{K}{(z-x)_{[d]}} + tS\sum_{I,J}\frac{a_{I,J}}{z_I^{J+1}(z-x)_{I^c}}\right)Qdz_{[d]} \tag{7.90}$$

$$= \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \frac{1}{(z-x)_{[d]}}\left(K + tS\sum_{I,J}\frac{a_{I,J}(z-x)_I}{z_I^{J+1}}\right)Qdz_{[d]} \tag{7.91}$$

$$= \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma R_0 Q dz_{[d]} \tag{7.92}$$

Now to turn this into a d-dimensional Fredholm problem we'll need to parametrize the integral. One way that might work is to set $z_k = r_k\exp(i\theta_k)$ and $x_k = s_k e^{i\phi_k}$. Since $dz_k = r_k i e^{i\theta_k}$ we have

$$(K+t\Gamma' S)Q = \frac{r_{[d]}}{(2\pi)^d}\int_0^{2\pi}\cdots\int_0^{2\pi} e^{i\theta_1}\ldots e^{i\theta_d}R_0 Q d\theta_{[d]} = q(\phi_1,\ldots,\phi_d) \tag{7.93}$$

so if we write

$$R = \exp(i(\theta_1+\ldots+\theta_d))R_0(\phi_1,\ldots,\phi_d,\theta_1,\ldots,\theta_d) \tag{7.94}$$

we have the inhomogeneous Fredholm equation of the first kind in d dimensions

$$\int_0^{2\pi} \cdots \int_0^{2\pi} RQd\theta_1 \ldots d\theta_d = g \tag{7.95}$$

where R is the kernel and $g = (2\pi)^d q/r_{[d]}$ and Q is in terms of θ_k's. Perhaps Fredholm theory can be applied in solving the random walk models in man dimensions. Perhaps this will also lead to a good inner product on the space where Q lives.

Chapter 8

Groups for walks with small steps

Consider $g_i(x_i) = X_i$ that fixes S where S is the Laurent step polynomial. We either have that g_i is trivial or

$$g_i(x_i) = \overline{x}_i \frac{S_i^{-1}}{S_i^1}, \quad g_i^2(x_i) = x_i \tag{8.1}$$

whenever $S_i^{\pm 1} \neq 0$. These bi-rational transformations generate a group G_S

$$G_S = \langle g_i | g_i^2 = 1 \rangle = \langle g_i | g_i^2 = 1 \text{ and } g_i \text{ is non-trivial} \rangle \tag{8.2}$$

where i ranges from 1 to d. When g_i is trivial the problem is essentially lower-dimensional

1) If $S_i^{-1} = 0$ and $S_i^1 = 0$, $S = S_i^0$ so the x_i variable plays no role since $E_i SQ = SQ$.

2) If $S_i^{-1} = 0$ and $S_i^1 \neq 0$ then

$$E_i SQ = SQ - \overline{x}_i (S_i^1 Q_i^{-1} + S_i^0 Q_i^{-1}) = SQ \tag{8.3}$$

   which is the same case as before

3) If $S_i^{-1} \neq 0$ but $S_i^1 = 0$ then $E_i SQ$ is non-trivial but since we cannot take any steps away from the $x_i = 0$ hyperplane, we see that S is essentially of lower

dimension.

This allows us to define the *dimension* of G_S or of the random walk model itself. Let $\dim(S)$ be the number of non-trivial generators g_i in this representation of the group G_S. Without loss of generality we will speak of models set in d dimensions that have a full set of generators. Firstly, we can easily show that there are infinitely many different group sizes that are manifested among the different dimensions. Let S be a $(d-1)$-dimensional model with group G of largest finite order (number of elements) for that dimension. Set this model in d dimensions and define a new model $S' = S(x_d + \bar{x}_d)$. This model has a non-trivial generator g_d that S lacks. Therefore the group G' of this new model satisfies

$$G' \cong G \times \langle g_d \rangle \tag{8.4}$$

and $|G'| = 2|G|$ and continuing in this manner allows us to write an infinite sequence of distinct groups. That is, not only are there infinitely many distinct groups but also, there is a group in d dimensions not encountered before in $(d-1)$ dimensions (neither in order nor structure). For example the orders of the finite groups set in 2 dimensions are 1,2,4,6 and 8. So in 3 dimensions we have groups of order 1,2,4,6,8,12 and 16 at least. In fact Steve's paper shows that there are even more - of orders 24 and 48 too.

8.1 Coxeter Groups

This discussion would be short without delving into the subject of Coxeter groups. The d-dimensional groups defined above are Coxeter groups - a generalization of the dihedral groups encountered in dimension 2 random walk models. Coxeter groups are defined as

$$G = \langle r_1, \ldots, r_d | (r_i r_j)^{m_{i,j}} = 1 \rangle \tag{8.5}$$

where $m_{i,i} = 1$ and $m_{i,j} \geq 2$ for all $i \neq j$, and $m_{i,j}$ could also be infinite. We can call these r's reflections. When $m_{i,j} = 2$ the reflections commute. Also $m_{i,j} = m_{j,i}$. The *rank* of G is the number of generators of in this specific representation of G. Often G will have more than one representation with different numbers of generators. So

in our discussion of groups for random walks with small steps the rank of G is the dimension of the model. We can draw a graph for each Coxeter group

- label vertices with the generators of G
- connect the vertices g_i and g_j if $m_{i,j} \geq 3$
- label the edge with $m_{i,j}$ if $m_{i,j} \geq 4$.

If the graph has different components, the Coxeter group factors as a direct product of Coxeter groups represented by the individual components. We can see this in terms of the step polynomial S. Suppose that S factors as

$$S = T(x_1, \ldots, x_k)R(x_{k+1}, \ldots, x_d) \tag{8.6}$$

For a fixed $1 \leq i \leq k$ we have

$$S_i^j = [x_i^j]S = R[x_i^j]T = RT_i^j \tag{8.7}$$

and therefore

$$g_i(x_i) = \overline{x}_i \frac{T_i^{-1}R}{T_i^1 R} = \overline{x}_i \frac{T_i^{-1}}{T_i^1} \tag{8.8}$$

Similarly if $k+1 \leq j \leq d$ then

$$g_j(x_j) = \overline{x}_j \frac{R_i^{-1}}{R_i^1} \tag{8.9}$$

Clearly

$$g_i g_j = g_j g_i \tag{8.10}$$

since they don't interfere with each other. Since i and j were arbitrary, this means that the generators from different sets commute and the Coxeter graph of the group factors into two pieces. We can write

$$G_S = G_T \times G_R \tag{8.11}$$

where the subscripts highlight which parts of S they groups are generated by. This factoring also allows us to write the generating function Q as a Hadamard product of lower-dimensional generating functions. We can call each component of the graph a prime model. We can say that a simple small component is one whose group doesn't split non-trivially (its graph has one component).

It's worth mentioning that it doesn't seem that all Coxeter groups are manifested in random walk models. Looking at the classification of finite Coxeter groups, those of rank 2 have orders 4, 6, 8, 12, 10 and $2p$ (manifested in a p-gon). Namely the groups of orders 10,12 and $2p$ (for most p) are not manifested in random walk models with small steps on the quarter plane. It appears that the p-gon symmetries are not present for most p because of the crystallographic nature of the problem. It's quite possible that the Crystallographic restriction theorem will apply to the random walk model groups in all dimensions. However, as can be seen, a lot of random walk model groups in 2 dimensions have generators that don't map the lattice back onto itself in a simple way because the transformations are bi-rational. We can define classes of groups.

- a model is "nice" iff $g_i(x_i) \in \mathbb{Z}[x_1, \overline{x}_1 \ldots, x_d, \overline{x}_d]$
- a model is "crystallographic" iff $g_i(x_1^{n_1} \ldots x_d^{n_d}) = x_1^{m_1} \ldots x_d^{m_d}$ for some integers n_j and m_j.

The names are suggestive because the generators in crystallographic random walk models map monomials to monomials and we can see if G is crystallographic. The next step up are "nice" models where monomials are mapped to Laurent polynomials. The rest of the classes have a generator mapping some monomials to rational functions. Perhaps those first two types of models can be represented via reflections and will thus have Weyl group.

8.2 Orbit Sums and Kernel Method

Since $G = G_S$ preserves the kernel K we can construct an orbit sum where we have

$$\sum_{g \in G} \text{sign}(g) g(x_1 \ldots x_d Q) = \frac{1}{K(x_1, \ldots, x_d)} \sum_{g \in G} \text{sign}(g) g(x_1 \ldots x_d) \tag{8.12}$$

We can write this equation more compactly via

$$D_G = \sum_{g \in G} \text{sign}(g), \quad D'_G = D_g x_1 \dots x_d \tag{8.13}$$

Then the identity reads in one of two followin ways

$$D'_G Q = \frac{1}{K} D'_G(1) \tag{8.14}$$

or

$$D'_G(KQ - 1) = 0 \tag{8.15}$$

We'll prove the identity shortly. Let g_k be one of the non-trivial generators. Then we can form the factor group $G/\langle g_k \rangle$ which naturally embeds into G as H. Since g_k is of order 2 H is normal in G and has two cosets - H and Hg_k. To make the discussion neater let e_g be the sign of g. This allows us to write D_G in another way

$$D_G = \sum_{g \in H} e_g g + \sum_{g \in H g_k} e_g g \tag{8.16}$$

$$= \sum_{g \in H} e_g g + \sum_{g \in H} (-1) e_g g g_k \tag{8.17}$$

$$= \left(\sum_{g \in H} e_g g \right) (1 - g_k) \qquad = D_H(1 - g_k) \tag{8.18}$$

We can prove the identity by starting with the unlabeled functional equation written as

$$KQ + \sum_{\emptyset \neq I \subseteq [d]} (-1)^{|I|} \bar{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} = 1 \tag{8.19}$$

We see that

$$D'_G(\bar{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}) = D_G(x_{[d] \setminus I} S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}) \tag{8.20}$$

Now, if we let $i \in I$ then $D_G = D_{H_i}(1 - g_i)$ and since there are no variables x_i in the expression D_G klls that term. Applying D'_G over the functional equation yields the

result.

For a dihedral group of order 4 we have

$$D_G = 1 - g_1 + g_2 g_1 - g_1 g_2 g_1 = (1 + g_2)(1 - g_1) \tag{8.21}$$

but another way to write it is by using the relation

$$g_1 g_2 g_1 g_2 = 1 \tag{8.22}$$

and multiplying both sides by g_2. We find that

$$D_G = 1 - g_1 - g_2 + g_2 g_1 = (1 - g_2)(1 - g_1) \tag{8.23}$$

For a dihedral group of order 6 we have

$$\begin{aligned} D_G &= 1 - g_1 + g_2 g_1 - g_1 g_2 g_1 + g_2 g_1 g_2 g_1 - g_1 g_2 g_1 g_2 g_1 && (8.24)\\ &= (1 + g_1 g_2 + (g_1 g_2)^2)(1 - g_1) && (8.25)\\ &= (1 + g_1 g_2 + g_2 g_1)(1 - g_1) && (8.26) \end{aligned}$$

The dihedral group of order 8 has

$$D_G = (1 + g_1 g_2 + g_2 g_1 + g_1 g_2 g_1 g_2)(1 - g_1) \tag{8.27}$$

This also works for the group generated by two elements within a larger Coxeter group. We have

$$D_{\langle g_i, g_j \rangle} = (1 + g_i g_j + \cdots + (g_i g_j)^{m_{i,j}-1})(1 - g_i) \tag{8.28}$$

where we can exchange the meaning of i and j to get the alternative form. These calculations show that we probably won't be able to factor more than one factor $(1 - g_k)$ out of the operator.

8.3 Half-Orbit Sums

Sometimes applying the operator D'_G to 1 gives 0. In the cases studied it seems to happen when we have $(x,y) \mapsto (y,x)$ as part of the group. Let $h = h_{i,j}$ be the map $(x_i,x_j) \mapsto (x_j,x_i)$. We can form the quotient group $G/\langle h \rangle$ as before and find H a subgroup of G where $H \cong G/\langle h \rangle$. Since h has order 2 we find that H is normal and $G = H \cup Hh$. If $\text{sign}(h) = -1$ we have

$$D_G = D_H(1-h), \quad (1-h)(x_1 \dots x_d) = 0 \tag{8.29}$$

and therefore this result is $D'_G(1) = 0$. This is an example where we would have to use the half-orbit method. But does this mean that every situation with $D'_G(1) = 0$ is a result of having an element like h within G. One can certainly imagine higher order permutations being present within G that can also lead to orbit of zero. Similarly, there may be group elements that are not permutations that also lead to orbit of zero. We can treat some of the possibilities as follows. Let $g \in G$ fix $x = x_1 \dots x_d$. That is

8.4 Infinite Group Orbits

For an infinite group of dimension 2 the orbit operator will perhaps have the following form

$$D_G = (1 + g + g^2 + \cdots)(1 - g_1) = (1-g)^{-1}(1-g_1) \tag{8.30}$$

where $g = g_1 g_2$ and we seek an inverse of $1-g$. However this will most likely be a difficult matter because g can be rather complicated in some cases.

Chapter 9

Orbit Summation for Walks with Interactions

The orbit summation method works well with unlabeled walk models with finite groups. It is a natural question to consider extending it to the labeled walk models. The problem is that it is no longer easy to eliminate the unknown functions. The functional equation in two dimensions is

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \alpha \tag{9.1}$$

The group $G = G_S$ has two non-trivial generators g_1 and g_2 and G has order 4, 6 or 8. We have the orbit summation operator D_G which interprets the orbit summation method as an alternating sum of group actions.

9.1 Group of order 4

From a previous section we found that

$$D_G = 1 - g_1 - g_2 + g_1g_2 = (1 - g_1)(1 - g_2) = \Delta_1\Delta_2 \tag{9.2}$$

where we define $\Delta_i = 1 - g_i$. In the unlabeled orbit summation method we had a direct application of D_G on the equations to perform the required cancellations. Here we will attempt to do something similar. Since the cancellations don't happen

automatically if we use D_G directly, we'll instead obtain a linear system of equations by applying the group operations on the original functional equation. Then we'll use linear algebra to find a solution where our variables will be the ones we wish to eliminate Q_i^0 and their images under the group actions. If the associated matrix has non-zero determinant we have a unique solution and unfortunately we won't be able to eliminate variables. However there are some cases where the determinant will be zero. Because of symmetry we expect to be able eliminate the desired variables. The group operations commute so we have

$$g_i Q_i^0 = Q_i^0, \quad g_i g_j Q_i^0 = g_j Q_i^0 \tag{9.3}$$

and therefore we have the variables that we wish to eliminate

$$E = \{Q_1^0, Q_2^0, g_2 Q_1^0, g_1 Q_2^0\}. \tag{9.4}$$

This is looking promising since we have four equations and four unknowns that we wish to eliminate. We re-write the original equation in a format centered on these variables

$$K_1 Q_1^0 + K_2 Q_2^0 = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha = \beta \tag{9.5}$$

For brevity let $g_{12} = g_1 g_2$

$$K_1 Q_1^0 + K_2 Q_2^0 = \beta \tag{9.6}$$

$$g_1 K_1 Q_1^0 + g_1 K_2 g_1 Q_2^0 = g_1 \beta = \beta_1 \tag{9.7}$$

$$g_2 K_1 g_2 Q_2^0 + g_2 K_2 Q_2^0 = g_2 \beta = \beta_2 \tag{9.8}$$

$$g_{12} K_1 g_2 Q_1^0 + g_{12} K_2 g_1 Q_2^0 = g_{12} \beta = \beta_{12} \tag{9.9}$$

$$\tag{9.10}$$

Using the displayed order of variables in E as our order for the system we can write the associated matrix M as

$$M_4 = \begin{bmatrix} K_1 & K_2 & 0 & 0 \\ g_1K_1 & 0 & 0 & g_1K_2 \\ 0 & g_2K_2 & g_2K_1 & 0 \\ 0 & 0 & g_{12}K_1 & g_{12}K_2 \end{bmatrix} \tag{9.11}$$

Its determinant is

$$\begin{aligned} \det M_4 &= K_1g_{12}K_1g_1K_2g_2K_2 - g_1K_1g_2K_1K_2g_{12}K_2 && (9.12)\\ &= K_1g_{12}K_1g_1K_2g_2K_2 - g_1(g_1K_1g_{12}K_1g_1K_2g_2K_2) && (9.13)\\ &= \Delta_1(K_1g_{12}K_1g_1K_2g_2K_2) && (9.14)\\ &= \Delta_2(g_1K_1g_2K_1K_2g_{12}K_2) && (9.15) \end{aligned}$$

where the last form comes from doing the process in the same way but with g_2 instead.

9.1.1 When is $\det M_4 = 0$?

Now our task is to find where this determinant is zero. Clearly the expression withing each Δ must be in the kernel of the operator. To that end let's find when that happens. For a Laurent polynomial f in the variables x_i and t we have

$$\Delta_k f = \sum_n f_n(1-g_k)x_k^n = \sum_n (f_n - f_{-n}\mu_k^{-n})x_i^n \tag{9.16}$$

where $\mu_k = S_k^{-1}/S_k^1$. This implies that

$$\begin{aligned} f_n &= f_{-n}\mu_k^{-n} && (9.17)\\ f_n(S_k^1)^n &= f_{-n}(S_k^{-1})^n && (9.18) \end{aligned}$$

for all $n \in \mathbb{Z}$. It's not clear if this will help us. However the models that exhibit groups of order 4 have at least one symmetry across a line - either vertical or

horizontal symmetry. WLOG let the model have vertical symmetry. That is

$$g_1(x_1) = x_1, \quad g_1(S_2^i) = S_2^i, \quad g_1(S_1^i) = S_1^{-i} \tag{9.19}$$

This allows us to compute some things. For example

$$g_1 K_2 = c_2 - t\bar{x}_2 g_1 S_2^{-1} = c_2 - t\bar{x}_2 S_2^{-1} = K_2 g_1 g_2 K_2 = g_2 g_1 K_2 = g_2 \tag{9.20}$$

Therefore Δ_1 fixes both K_2 and $g_2 K_2$. We see that the determinant is

$$\det M_4 = \Delta_1(K_1 K_2 g_{12} K_1 g_2 K_2) = K_2 g_2 K_2 \Delta_1(K_1 g_{12} K_1) \tag{9.21}$$

This is zero if and only if $\Delta_1(K_1 g_{12} K_1) = 0$ so we have

$$\begin{aligned}
0 &= \Delta_1(K_1 g_{12} K_1) && (9.22)\\
&= K_1 g_{12} K_1 - g_1 K_1 g_2 K_2 && (9.23)\\
&= (c_1 - t\overline{x_1} S_1^{-1})(c_1 - t x_1 g_2 S_1^{-1}) - (c_1 - t x_1 S_1^{-1})(c_1 - t\overline{x_1} g_2 S_1^{-1}) && (9.24)\\
&= c_1 t\bar{x}_1(g_2 S_2^{-1} - S_2^{-1}) - c_1 t x_1(g_2 S_2^{-1} - S_2^{-1}) && (9.25)\\
&= c_1 t(\bar{x}_1 - x_1)(g_2 S_1^{-1} - S_1^{-1}) && (9.26)\\
&= -c_1 t(\bar{x}_1 - x_1)\Delta_2(S_1^{-1}) && (9.27)
\end{aligned}$$

This is true if and only if

$$g_2 S_1^{-1} = S_1^{-1} \tag{9.28}$$

We have

$$g_2(S_{1,2}^{-1,-1}\bar{x}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1} x_2) = S_{1,2}^{-1,-1} x_2 \overline{\mu}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}\bar{x}_2 \mu_2 \tag{9.29}$$

and

$$S_{1,2}^{-1,-1} x_2 \overline{\mu}_2 + S_{1,2}^{-1,1}\bar{x}_2 \mu_2 = S_{1,2}^{-1,-1}\bar{x}_2 + S_{1,2}^{-1,1} x_2 \tag{9.30}$$

which can be re-written succinctly as

$$S_{1,2}^{-1,-1} = \mu_2 S_{-1,1} \tag{9.31}$$

If one of the S's is zero so is the other and therefore by symmetry the model doesn't have diagonal steps. Since we have two non-trivial generators this implies that both x_2 and $\bar{x}_2$ are present in S, so the model trivially exhibits a symmetry about the x_1 axis. If the S's are non-zero this implies μ_2 is a constant and therefore $S_2^{-1} = S_2^1$. Again the model exhibits symmetry about the x_1 axis. This leads to a theorem of sorts. The determinant of M_4 is zero if and only if the model is highly-symmetric. These are the only cases where we have hope that the orbit summation method will extend to the labeled cases. Note that in the unlabeled case $c_1, c_2 = 0$ so the analysis yields a zero determinant in any group 4 case, confirming that we can use the orbit summation method.

9.1.2 Analogue of the Orbit Summation Method

We are now ready to attempt using the orbit summation method in the highly symmetric cases. There are 8 such models but only 5 of them have dimension 2. Furthermore 2 of the models are reflections of each other so we are really left with 4 interesting cases - diagonal walks, full walks, full walks minus two horizontal steps. If we are able to eliminate all the variables then we'll be left with a linear relation

$$\gamma\beta - \gamma_1\beta_1 - \gamma_2\beta_2 + \gamma_{12}\beta_{12} = 0 \tag{9.32}$$

where the choice of signs will make things neater later. Let $\bullet = g_1$ and $* = g_2$. First we have $K_2^{\bullet} = K_2$ and $K_1^{*} = K_1$. Then we let

$$x = Q_1^0, y = Q_2^0, u = Q_1^{0,*}, v = Q_2^{0,\bullet} \tag{9.33}$$

and

$$a = K_1, b = K_1^{\bullet}, c = K_2, d = K_2^{*} \tag{9.34}$$

Then the equations from the system can be described as

$$\beta = ax + cy, \quad \beta_1 = bx + cv \tag{9.35}$$

$$\beta_2 = dy + au, \quad \beta_{12} = bu + dv \tag{9.36}$$

Therefore the above equation needs to satisfy

$$0 = (ax+cy)\gamma - (bx+cv)\gamma_1 - (dy+au)\gamma_2 + (bu+dv)\gamma_{12} \tag{9.37}$$

$$= (a\gamma - b\gamma_1)x + (c\gamma - d\gamma_2)y + (b\gamma_{12} - a\gamma_2)u + (d\gamma_{12} - c\gamma_1)v \tag{9.38}$$

and in order to eliminate the variables x, y, u, v we need to make all their coefficients zero. Therefore

$$a\gamma = b\gamma_1, \quad c\gamma = d\gamma_2, \quad b\gamma_{12} = a\gamma_2, \quad d\gamma_{12} = c\gamma_1 \tag{9.39}$$

If we set $\gamma = 1$ then $\gamma_1 = a/b$, $\gamma_2 = c/d$ and $\gamma_{12} = (ac)/(bd)$. So we can multiply out by bd to find the relation

$$bd\beta - ad\beta_1 - cb\beta_2 + ac\beta_{12} = 0 \tag{9.40}$$

where we have successfully eliminated $Q_1^0, Q_2^0, Q_1^{0,*}$, and $Q_2^{0,\bullet}$. Or written in terms of the K's we have

$$K_1^\bullet K_2^* \beta - K_1 K_2^* \beta^\bullet - K_1^\bullet K_2 \beta^* + K_1 K_2 \beta^{\bullet *} = 0 \tag{9.41}$$

Written in operator notation this equation becomes much nicer

$$(1-\bullet)(1-*)(K_1^\bullet K_2^* \beta) = \Delta(K_1^\bullet K_2^* \beta) = 0 \tag{9.42}$$

Let $K_0 = K_1^\bullet K_2^*$. We can now write this equation in another way

$$\Delta(K_0\beta) = \Delta(K_0(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) \tag{9.43}$$

$$= \Delta(K_0 KQ) + \Delta(K_0 K_{1,2} Q_{1,2}^{0,0}) - \Delta(K_0\alpha) \tag{9.44}$$

$$= K\Delta(K_0 Q) + \Delta(K_0 K_{1,2}) Q_{1,2}^{0,0} - \Delta(K_0)\alpha \tag{9.45}$$

where the last line follows from Δ fixing K and functions without variables x_i. All that remains is to calculate the various quantities.

$$\Delta g_i K_i = -\Delta K_i = -(c_i - t x_i S_i^{-1} - c_i + t\overline{x}_i S_i^{-1}) = -t(\overline{x}_i - x_i) S_i^{-1} \tag{9.46}$$

Therefore

$$\Delta K_0 = \Delta_1(K_1^\bullet)\Delta_2(K_2^*) = t^2(\overline{x}_1 - x_1)(\overline{x}_2 - x_2) S_1^{-1} S_2^{-1} \tag{9.47}$$

Next

$$\Delta(K_0 K_{1,2}) = \Delta_1 K_1^\bullet \Delta_2 K_2^* K_{1,2} = \Delta_1 K_1^\bullet (K_2^* K_{1,2} - K_2 K_{1,2}^*) \tag{9.48}$$

We calculate the quantity in parentheses as

$$K_2^* K_{1,2} - K_2 K_{1,2}^* = \tag{9.49}$$
$$= (c_2 - t x_2 S_2^{-1})(c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) - (c_2 - t\overline{x_2} S_2^{-1})(c_1 c_2 - t\overline{x_1} x_2 S_{1,2}^{-1,-1}) \tag{9.50}$$
$$= c_1 c_2 t S_2^{-1}(\overline{x_2} - x_2) - c_2 t\overline{x_2} S_{1,2}^{-1,-1}(\overline{x_2} - x_2) \tag{9.51}$$
$$= c_2 t(\overline{x_2} - x_2)(c_1 S_2^{-1} - \overline{x_1} S_{1,2}^{-1,-1}) \tag{9.52}$$

and thus

$$\Delta(K_0 K_{1,2}) = c_2 t(\overline{x_2} - x_2)\Delta_1 K_1^\bullet (c_1 S_2^{-1} - \overline{x_1} S_{1,2}^{-1,-1}) \tag{9.53}$$
$$= c_2 t(\overline{x_2} - x_2)\Delta_1 K_1^\bullet (c_1 S_2^{-1} - \overline{x_1} S_{1,2}^{-1,-1}) \tag{9.54}$$
$$\tag{9.55}$$

To compute this note that

$$\Delta_1 K_1^\bullet c_1 S_2^{-1} = c_1 S_2^{-1} \Delta_1 K_1^\bullet = -c_1 t S_2^{-1}(\overline{x}_1 - x_1) \tag{9.56}$$

and

$$\Delta_1 K_1^\bullet \overline{x_1} S_{1,2}^{-1,-1} = S_{1,2}^{-1,-1} \Delta_1 K_1^\bullet \overline{x_1} = c_1(\overline{x}_1 - x_1) S_{1,2}^{-1,-1} \tag{9.57}$$

Finally

$$\Delta(K_0K_{1,2}) = c_1c_2t(\bar{x}_1 - x_1)(\bar{x}_2 - x_2)(tS_1^{-1}S_2^{-2} - S_{1,2}^{-1,-1}). \tag{9.58}$$

Combining terms and manipulating things a bit leads us to the equation

$$\Delta(K_0Q) = \alpha t(\bar{x}_1 - x_1)(\bar{x}_2 - x_2)(tS_1^{-1}S_2^{-1} - b_1b_2(tS_1^{-1}S_2^{-1} - S_{1,2}^{-1,-1})Q_{1,2}^{0,0}) \tag{9.59}$$

or

$$\Delta(K_0Q) = \alpha t(\bar{x}_1 - x_1)(\bar{x}_2 - x_2)(tS_1^{-1}S_2^{-1}(1 - b_1b_2Q_{1,2}^{0,0}) + S_{1,2}^{-1,-1}Q_{1,2}^{0,0}) \tag{9.60}$$

9.1.3 Similarities between labeled and unlabeled versions

We start with the observation that $\Delta \equiv D_G$ for the case of highly-symmetric random walk models

$$D_G(K_1^\bullet K_2^*(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) = 0 \tag{9.61}$$

If we move down the the unlabelled case (assuming we can do with through this equation) we have

$$K_1^\bullet \mapsto -tx_1S_1^{-1} \tag{9.62}$$

$$K_2^* \mapsto -tx_2S_2^{-1} \tag{9.63}$$

$$K_{1,2} \mapsto -t\bar{x}_1\bar{x}_2S_{1,2}^{-1,-1} \tag{9.64}$$

and the equation becomes

$$t^2S_1^{-1}S_2^{-1}D_G(x_1x_2(KQ - t\bar{x}_1\bar{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{0,0} - 1) \tag{9.65}$$

since Δ fixes S_i^{-1}. The $Q_{1,2}^{0,0}$ quantity disappears after the action of D_G and we arrive at

$$D_G'(KQ - 1) = 0 \tag{9.66}$$

which was the orbit sum equation.

9.2 Group of order 6

Here we have some interactions as well but the problem is harder. We still have that $g_i Q_i^0 = Q_i^0$. We need to see what kind of expressions we get from the Q_i^0. Here we tabulate the sequences of operations g_1 and g_2 and which parts of those sequences survive application on Q_1^0, Q_2^0, K_1 and K_2. For example

$$g_{2121} Q_1^0 = g_{212} Q_1^0 = g_{121} Q_1^0 = g_{12} Q_1^0 \tag{9.67}$$

so the table has 12 in the appropriate rown/column.

Sequence	Reverse	Q1	Q2	K1	K2
1	21212	n/a	1	1	1
21	1212	2	21	21	21
121	212	12	21	121	212
2121	12	12	1	12	12
12121	2	2	n/a	2	2

As the talbe shows we really have only 6 variables. Let

$$u_1 = Q_1^0, \quad u_2 = g_2 Q_1^0, \quad u_3 = g_{12} Q_1^0 \tag{9.68}$$

$$v_1 = Q_2^0, \quad v_2 = g_1 Q_2^0, \quad v_3 = g_{21} Q_2^0 \tag{9.69}$$

$$\tag{9.70}$$

Also there are 10 different K_1's and K_2's which we label sequentially as $A_1, \ldots, A_6$ and $B_1, \ldots, B_6$ respectively. We have the associated 6×6 matrix

$$M_6 = \begin{bmatrix} A_1 & & & B_1 & & \\ A_2 & & & & B_2 & \\ & A_3 & & & & B_3 \\ & & A_4 & & & B_4 \\ & & A_5 & & B_5 & \\ & A_6 & & B_6 & & \end{bmatrix} \tag{9.71}$$

with determinant

$$\det M_6 = A_2A_4A_6B_1B_3B_5 - A_1A_3A_5B_2B_4B_6 \tag{9.72}$$

What is nice about these coefficients is that $A_1 = g_1A_2$ and so on and thus we can write

$$\det M_6 = \Delta_1(A_2A_4A_6B_1B_3B_5) = \Delta_1(g_1K_1g_{121}K_1g_{12}K_1K_2g_{21}K_2g_{12}K_2) \tag{9.73}$$

We can easily check the relationships A's and B's in the two group cases. For the models with the first group

- $S = x_1\bar{x}_2 + \bar{x}_1 + x_2 : A_1 = A_6, A_2 = A_5, A_3 = A_4$ and $B_1 = B_4, B_2 = B_3, B_5 = B_6$.
- $S = \bar{x}_1x_2 + x_1 + \bar{x}_2$: $A_1 = A_4, A_2 = A_3, A_5 = A_6$ and $B_1 = B_2, B_3 = B_6, B_4 = B_5$
- $S = x_1 + \bar{x}_1 + x_2 + \bar{x}_2 + x_1\bar{x}_2 + \bar{x}_1x_2$: We have no non-trivial relations.

For the models with the second group we have

- $S = x_1x_2 + \bar{x}_1 + \bar{x}_2 : A_1 = A_6, A_2 = A_5, A_3 = A_4$ and $B_1 = B_2, B_3 = B_6, B_4 = B_5$
- $S = \bar{x}_1\bar{x}_2 + x_1 + x_2$: $A_1 = A_4, A_2 = A_3, A_5 = A_6$ and $B_1 = B_4, B_2 = B_3, B_5 = B_6$.
- $S = x_1 + \bar{x}_1 + x_2 + \bar{x}_2 + x_1x_2 + \bar{x}_1\bar{x}_2$: We have no non-trivial relations.

For the cases with relations it is clear that $\det M_6 = 0$. Using a computer we can easily check that the determinant is not zero in the cases without nontrivial relations. It's worth mentioning that so far all groups that have had determinant zero have been crystallographic but as we can see not all crystallographic groups end up having determinant zero.

We seek the same type of relation on β and its orbit as before so we'll use D_G as inspiration. We have the following

$$\begin{aligned} 0 =& \gamma\beta - \gamma_1\beta_1 + \gamma_{21}\beta_{21} - \gamma_{121}\beta_{121} + \gamma_{12}\beta_{12} - \gamma_2\beta_2 && (9.74)\\ =& \gamma(A_1u_1 + B_1v_1) - \gamma_2(A_6u_2 + B_6v_1) - \gamma_1A_2u_1 + B_2v_2) && (9.75)\\ &+ \gamma_{12}(A_5u_3 + B_5v_2) + \gamma_{21}(A_3u_2 + B_3v_3) - \gamma_{121}(A_4u_3 + B_4v_3). && (9.76) \end{aligned}$$

Expanding out and collecting terms leads to the following equations

$$A_1\gamma = A_2\gamma_1, \quad A_3\gamma_{21} = A_6\gamma_2, \quad A_5\gamma_{12} = A_4\gamma_{121} \tag{9.77}$$

$$B_1\gamma = B_6\gamma_2, \quad B_5\gamma_{12} = B_2\gamma_1, \quad B_3\gamma_{21} = B_4\gamma_{121} \tag{9.78}$$

This system of equations doesn't have a nontrivial symbolic solution so we need to use the relations here. There is a rational solution in each case with relations where γ is a free variable. We set γ to be the least common multiple of the denominators. In the six cases we've performed the calculations and checked that indeed $K_0 = \gamma$ in each case.

- $S = x_1\bar{x}_2 + \bar{x}_1 + x_2 : \quad \gamma = K_0 = A_3A_5B_6 = g_{21}(K_1)g_{12}(K_1)g_2(K_2)$
- $S = \bar{x}_1x_2 + x_1 + \bar{x}_2: \quad \gamma = K_0 = A_2A_5B_6 = g_1(K_1)g_{12}(K_1)g_2(K_2)$
- $S = x_1x_2 + \bar{x}_1 + \bar{x}_2 : \quad \gamma = K_0 = A_2A_3B_5B_6 = g_1(K_1)g_{21}(K_1)g_{12}(K_2)g_2(K_2)$
- $S = \bar{x}_1\bar{x}_2 + x_1 + x_2: \quad \gamma = K_0 = A_2B_6 = g_1(K_1)g_2(K_2)$

Here we've picked the respresentations of A's and B' with the least number of group operations g_i to reach the resulting expression. We also tried finding a possible K_0 for the cases with no relations but we didn't find any among products of distinct A's and B's.

9.3 Group of order 8

The analysis for groups of order 8 is quite similar so we shall omit it. Here are the results. The two models are

$$S = x_1 + \bar{x}_1 + x_1\bar{x}_2 + \bar{x}_1x_2 \qquad S = x_1 + \bar{x}_1 + x_1x_2 + \bar{x}_1\bar{x}_2 \tag{9.79}$$

The relations are the same in both models

$$A_1 = A_6, A_2 = A_5, A_3 = A_4, A_7 = A_8 \tag{9.80}$$

$$B_1 = B_4, B_2 = B_3, B_4 = B_8, B_6 = B_7 \tag{9.81}$$

In both cases

$$\gamma = K_0 = A_2A_3B_5B_6 = g_1(K_1)g_{21}(K_1)g_{212}(K_2)g_{1212}(K_2) \tag{9.82}$$

Here's we've also written them in their shortest forms.

9.4 Infinite Groups

A natural question is whether we can extend this analysis to handle infinite groups, at least in a purely formal sense. Here the orbits will be infinitely long and while there will be relations between the gQ_i^0 variables from one orbit iteration to the next, there will be no point where we can stop that we'll have as many equations as variables - there will always be more variables. Thus we'll need to take an infinite summation of the equations. It's quite possible that any candidates for K_0 will be infinite products. Since the K_i's have t^0 coefficients, such products will not converge. However, there may be hope if there are enough relations between the gK_i's so that we may have a finite product of gK_i's for K_0. The unlabelled orbit method doesn't need a K_0 to get rid of all terms Q_i^0 and $Q_{1,2}^{0,0}$. One would hope that something like this would work for the infinite case.

Chapter 10

Highly-symmetric 2D models

When we consider the highly-symmetric cases the x-transformations are given by

$$\bullet : x_1 \mapsto \overline{x_1}, \quad * : x_2 \mapsto \overline{x_2} \tag{10.1}$$

In each case we have

$$K^{\bullet} = K^{*} = K, \quad K_2^{\bullet} = K_2, \quad K_1^{*} = K_1 \tag{10.2}$$

while the others change in some way. The solutions eliminating the kernel also have a nice form

$$\sigma_i = \frac{1 - S_i^0 - \sqrt{(1-S_i^0)^2 - 4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1 - S_i^0 + \sqrt{(1-S_i^0)^2 - 4t^2(S_i^1)^2}}{2tS_i^1} \tag{10.3}$$

because we have

$$\overline{\sigma}_i = \frac{S_i^1}{S_i^{-1}} conj(\sigma_i) = conj(\sigma_i) \tag{10.4}$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0 \tag{10.5}$$
$$-KQ^* + K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0 \tag{10.6}$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0 \tag{10.7}$$
$$K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0 \tag{10.8}$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1 K_2^* Q_1^0 + K_2^* K_2 Q_2^0 - K_2^* K_{1,2} Q_{1,2}^{0,0} + \alpha K_2^* = 0 \tag{10.9}$$
$$K_1 K_2 Q_1^{0,*} + K_2 K_2^* Q_2^0 - K_2 K_{1,2}^* Q_{1,2}^{0,0} + \alpha K_2 = 0 \tag{10.10}$$

We subtract these two equations to obtain

$$K_1 (K_2^* Q_1 - K_2 Q_1^*) - (K_2^* K_{1,2} - K_2 K_{1,2}^*) Q_{1,2} + \alpha (K_2^* - K_2) = 0 \tag{10.11}$$
$$K_2^* Q_1 - K_2 Q_1^* - \frac{K_2^* K_{1,2} - K_2 K_{1,2}^*}{K_1} Q_{1,2} + \alpha \frac{K_2^* - K_2}{K_1} = 0 \tag{10.12}$$
$$\tag{10.13}$$

The differences in solutions will occur from here on out. We have

$$K_2^* - K_2 = c_2 - t x_2 S_2^{-1} - c_2 + t \overline{x_2} S_2^{-1} = t(\overline{x_2} - x_2) S_2^{-1}(\sigma_1) \tag{10.14}$$

Next

$$K_2^* K_{1,2} - K_2 K_{1,2}^* = (c_2 - t x_2 S_2^{-1})(c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) \tag{10.15}$$

$$- (c_2 - t\overline{x_2} S_2^{-1})(c_1 c_2 - t\overline{x_1} x_2 S_{1,2}^{-1,-1}) \tag{10.16}$$

$$= c_1 c_2 t S_2^{-1}(\overline{x_2} - x_2) - c_2 t \overline{x_2} S_{1,2}^{-1,-1}(\overline{x_2} - x_2) \tag{10.17}$$

$$= c_2 t(\overline{x_2} - x_2)(c_1 S_2^{-1}(\sigma_1) - \overline{x_1} S_{1,2}^{-1,-1}) \tag{10.18}$$

10.1 Cartesian model: steps N, S, E, W

In this case $K_{1,2} = c_1 c_2$ so we have a simpler situation where we can re-write the above equation as

$$K_2^* Q_1 - K_2 Q_1^* + \alpha\beta \frac{K_2^* - K_2}{K_1} = 0 \tag{10.19}$$

where

$$\beta = 1 - b_1 b_2 Q_{1,2}^{0,0} \tag{10.20}$$

Furthermore

$$K_2^* - K_2 = t(\overline{x_2} - x_2) \tag{10.21}$$

So we can let

$$f = f(c_1, x_2) = \frac{t(\overline{x_2} - x_2}{K_1} \tag{10.22}$$

since K_1 doesn't depend on c_2 or x_1. The equation finally becomes

$$(c_2 - t x_2) Q_1^0(x_2) - (c_2 - t\overline{x_2}) Q_1^0(\overline{x_2}) + \alpha\beta f = 0 \tag{10.23}$$

We perform the extraction $[x_2^1]$ to get

$$c_2 Q_{1,2}^{0,1} - t Q_{1,2}^{0,0} + \alpha\beta f_2^1 = 0 \tag{10.24}$$

and the extraction $[b_2^0]$ to get

$$-t\widehat{Q}_{1,2}^{0,0} + \overline{a_1} f_2^1 \tag{10.25}$$

Using the second equation within the first one and simplifying yields

$$b_2 Q_{1,2}^{0,1} = t(a_2 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0}) \tag{10.26}$$
$$b_1 Q_{1,2}^{1,0} = t(a_1 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0}) \tag{10.27}$$

Here the second equation is obtained via the same method but with $\bullet$ instead of $*$. Using the same method we can also obtain

$$b_2 Q_{1,2}^{1,1} = t(a_2 Q_{1,2}^{1,0} - \beta \widehat{Q}_{1,2}^{1,0}) \tag{10.28}$$

Multiplying by b_1 and using the above relations leads to

$$b_1 b_2 Q_{1,2}^{1,1} = t(a_2 b_1 Q_{1,2}^{1,0} - \beta b_1 \widehat{Q}_{1,2}^{1,0}) \tag{10.29}$$
$$= t a_2 t(a_1 Q_{1,2}^{0,0} - \beta \widehat{Q}_{1,2}^{0,0}) - t\beta t(a_1 \widehat{Q}_{1,2}^{0,0} - \beta \widehat{\widehat{Q}}_{1,2}^{0,0}) \tag{10.30}$$
$$= t^2(a_1 a_2 Q_{1,2}^{0,0} - a_2 \beta \widehat{Q}_{1,2}^{0,0} - a_1 \beta \widehat{Q}_{1,2}^{0,0} + \beta^2 \widehat{\widehat{Q}}_{1,2}^{0,0}) \tag{10.31}$$

Let

$$G = Q_{1,2}^{0,0}, \quad \widehat{G} = G_1, \quad \widehat{G} = G_2 \tag{10.32}$$

From before we have

$$Q_{1,2}^{0,0} = 1 + t a_1 a_2 (Q_{1,2}^{1,0} + Q_{1,2}^{0,1}) \tag{10.33}$$

Multiplying by $b_1 b_2$ and using the above relations leads to

$$b_1 b_2 G = b_1 b_2 + t a_1 a_2 (b_2 b_1 Q_{1,2}^{1,0} + b_1 b_2 Q_{1,2}^{0,1}) \tag{10.34}$$
$$= b_1 b_2 + t a_1 a_2 (b_2 t(a_1 G - \beta G_1) + b_1 t(a_2 G - \beta G_2) \tag{10.35}$$
$$= b_1 b_2 + t^2 a_1 a_2 (b_2 a_1 G + b_1 a_2 G - b_2 \beta G_1 - b_1 \beta G_2) \tag{10.36}$$

Using the relation for β and simplifying leads

$$G = \frac{b_1 b - 2 - t^2 a_1 a_2 (b_2 G_1 + b_1 G_2)}{b_1 b_2 - t^2 a_1 a_2 (a_2 b_1 + a_1 b_2 + b_1 b_2 (b_2 G_1 + b_1 G_2))} \tag{10.37}$$

If we let $F = (b_2 G_1 + b_1 G_2)$ and perform the long division with respect to F we find that

$$Q_{1,2}^{0,0} = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right) \tag{10.38}$$

Using arguments similar to those in the later section on diagonal steps we can find out more about the various Q's.

$$b_2 Q_{1,2}^{m,n+1} = t(a_2 Q_{1,2}^{m,n} - \beta \widehat{Q}_{1,2}^{m,n}) \tag{10.39}$$

$$b_1 Q_{1,2}^{m+1,n} = t(a_1 Q_{1,2}^{m,n} - \beta \widehat{Q}_{1,2}^{m,n}) \tag{10.40}$$

Summing the equations multiplied by their respective powers of x_1 and x_2 we find the equations

$$b_2 Q_2^{n+1} = t(a_2 Q_2^n - \beta \widehat{Q}_2^n) \tag{10.41}$$

$$b_1 Q_1^{m+1} = t(a_1 Q_1^m - \beta \widehat{Q}_1^m) \tag{10.42}$$

Now we can sum with respect to x_2 and x_1 to find

$$b_2 (Q - Q_2^0) = x_2 t(a_2 Q - \beta \widehat{Q}) \tag{10.43}$$

$$b_1 (Q - Q_1^0) = x_1 t(a_1 Q - \beta \widehat{Q}) \tag{10.44}$$

We can solve these for Q_2^0 and Q_1^0 as follows

$$b_2 Q_2^0 = (b_2 - a_2 x_2 t) Q + x_2 t \beta \widehat{Q} \tag{10.45}$$

$$b_1 Q_1^0 = (b_1 - a_1 x_1 t) Q + x_1 t \beta \widehat{Q} \tag{10.46}$$

$$\tag{10.47}$$

We can substitute these two expressions into the original functional equation and after some algebra we find that

$$Q = \frac{(1-b_1b_2G)(\alpha + tx_1\overline{b_1}K_1\widehat{Q} + tx_2\overline{b_2}K_2\widehat{Q})}{K-(1-\overline{c_1}x_1t)K_1-(1-\overline{c_2}x_2t)K_2} \tag{10.48}$$

and after recognizing some of the quantities as modified K_1 and K_2's we finally stop at the solution for Q

$$Q = \frac{(1-b_1b_2G)(\alpha - \overline{b}_1K_1\widehat{K}_1^{\bullet}\widehat{Q} - \overline{b}_2K_2\widehat{K}_2^{*}\widehat{Q})}{K-\overline{c}_1K_1K_1^{\bullet}-\overline{c}_2K_2K_2^{*}} \tag{10.49}$$

This expression is as perplexing as it is complicated. Why should the K's be combined in such a ridiculous way in order to give Q in terms of

10.2 Diagonal model: steps NE, NW, SE, SW

In this case the roots of the kernel become

$$\sigma_i = \frac{1-\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1+\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1} \tag{10.50}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}. \tag{10.51}$$

First we have

$$K_2^* - K_2 = t(\overline{x_2}-x_2)S_2^{-1}(\sigma_1) \tag{10.52}$$

$$= t(\overline{x_2}-x_2)(\overline{\sigma_1}+\sigma_1) \tag{10.53}$$

$$= \frac{\overline{x_2}-x_2}{\overline{x_2}+x_2} \tag{10.54}$$

We have

$$K_2^* K_{1,2} - K_2 K_{1,2}^* = (c_2 - t x_2 S_2^{-1})(c_1 c_2 - t\overline{\sigma_1}\overline{x_2}) - (c_2 - t\overline{x_2} S_2^{-1})(c_1 c_2 - t\overline{\sigma_1} x_2) \tag{10.55}$$

$$= c_1 c_2 t(\overline{x_2} - x_2) S_2^{-1} - c_2 t\overline{\sigma_1}(\overline{x_2} - x_2) \tag{10.56}$$

$$= c_2 t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \tag{10.57}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1 - \sqrt{1 - 4t^2 v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1 + \sqrt{1 - 4t^2 v^2}}{2tv} \tag{10.58}$$

We have

$$(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) = \frac{c1}{tv} - \frac{1 + \sqrt{1 - 4t^2 v^2}}{2tv} \tag{10.59}$$

$$= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2 v^2}}{2}\right) \tag{10.60}$$

Also

$$K_1 = c_1 - tv\overline{\sigma_1} \tag{10.61}$$

$$= c_1 - tv\frac{1 + \sqrt{1 - 4t^2 v^2}}{2tv} \tag{10.62}$$

$$= 1 - \overline{a_1} - \frac{1 + \sqrt{1 - 4t^2 v^2}}{2} \tag{10.63}$$

$$= -\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2 v^2}}{2} \tag{10.64}$$

Thus we finally have

$$\frac{1}{K_1}(K_2^* K_{1,2} - K_2 K_{1,2}^*) = -\frac{c_2 t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1})}{c_1 - t\overline{\sigma_1} v} \tag{10.65}$$

$$= -\frac{c_2 t(\overline{x_2} - x_2)(-1)}{tv} \tag{10.66}$$

$$= \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} \tag{10.67}$$

Thus the equation will now read as

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} Q_{1,2} + \alpha \frac{\overline{x_2} - x_2}{(\overline{x_2} + x_2) K_1} = 0 \tag{10.68}$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2} + x_2$. We have

$$R^* = (\overline{x_2} + x_2) K_2^* = c_2(\overline{x_2} + x_2) - x_2 = c_2 \overline{x_2} - \overline{a_2} x_2 \tag{10.69}$$

$$R = (\overline{x_2} + x_2) K_2 = c_2(\overline{x_2} + x_2) - \overline{x_2} = c_2 x_2 - \overline{a_2 x_2} \tag{10.70}$$

$$\tag{10.71}$$

We have a couple of relations here

$$R^* - R = v(K_2^* - K_2) = \overline{x_2} - x_2 \tag{10.72}$$

$$\overline{x_2} R^* - x_2 R = \overline{x_2} c_2 v - 1 - x_2 c_2 v + 1 \tag{10.73}$$

$$= c_2 v(\overline{x_2} - x_2) \tag{10.74}$$

The equation becomes

$$(c_2 \overline{x_2} - \overline{a_2} x_2) Q_1^0(x_2) + (\overline{a_2 x_2} - c_2 x_2) Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + (\overline{x_2} - x_2) \overline{K}_1 \alpha = 0 \tag{10.75}$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2} \delta_0 = 0 \tag{10.76}$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1} S_1^{-1}) Q_1^0 - (c_2 - t\overline{x_2} S_2^{-1}) Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) Q_{1,2}^{0,0} = \overline{a_1 a_2} \tag{10.77}$$

to further our work here. For diagonal steps it is

$$(1-tS)Q-(c_1-tv\overline{x_1})Q_1^0-(c_2-t\overline{x_2}S_2^{-1})Q_2^0+(c_1c_2-t\overline{x_1x_2})Q_{1,2}^{0,0}=\overline{a_1a_2} \tag{10.78}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0=\overline{a_2}+c_2Q_{1,2}^{0,0}+a_1tvQ_1^1 \tag{10.79}$$

We can extract higher coefficients too. Let $k\geq 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned}0&=Q_1^k-tS_1^{-1}Q_1^{k+1}-tS_1^0Q_1^k-tS_1^1Q_1^{k-1}-c_2Q_{1,2}^{k,0} && (10.80)\\ &\quad+t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0}+S_{1,2}^{0,-1}Q_{1,2}^{k,0}+S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) && (10.81)\\ &=Q_1^k-tvQ_1^{k+1}-tvQ_1^{k-1}-c_2Q_{1,2}^{k,0}+t\overline{x_2}(Q_{1,2}^{k+1,0}+Q_{1,2}^{k-1,0}) && (10.82)\end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k=tv(Q_1^{k+1}+Q_1^{k-1})+c_2Q_{1,2}^{k,0}-t\overline{x_2}(Q_{1,2}^{k+1,0}+Q_{1,2}^{k-1,0}) \tag{10.83}$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$\begin{aligned}Q_1^0&=\overline{a_2}+c_2Q_{1,2}^{0,0}+a_1tvQ_1^1 && (10.84)\\ Q_1^{0,*}&=\overline{a_2}+c_2Q_{1,2}^{0,0}+a_1tvQ_1^{1,*} && (10.85)\\ & && (10.86)\end{aligned}$$

Combining them with equation (10.76) gives

$$
\begin{aligned}
0 &= R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 && (10.87)\\
&= R^*(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) && (10.88)\\
&\quad - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 && (10.89)\\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2 Q_{1,2}^{0,0}) && (10.90)\\
&\quad - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 && (10.91)\\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 && (10.92)\\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0) && (10.93)
\end{aligned}
$$

So we can write the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a_2}\delta_1 = 0 \tag{10.94}$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1 tv} \tag{10.95}$$

This equation together with equation (10.76) form the base cases for the following lemma

Lemma 37. *Let $m \geq 0$ be an integer. Then*

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{10.96}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1,0)$. Now we assume that the formula is true for all cases up to

and including m. Then we use formula (10.83)

$$
\begin{aligned}
& R^* Q_1^m - R Q_1^{m,*} && (10.97)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(R^* Q_1^{m-1} - R Q_1^{m-1,*}) && (10.98)\\
& + c_2(R^* - R) Q_{1,2}^{m,0} - t(\overline{x_2} R^* - x_2 R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) && (10.99)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2) Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1}) && (10.100)\\
& + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} - c_2 tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) && (10.101)\\
=& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} && (10.102)\\
& - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0}) - \overline{a_2} tv\delta_{m-1}) && (10.103)
\end{aligned}
$$

Combining this with equation (10.96) leads to

$$
\begin{aligned}
0 &= tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} - \overline{a_2} tv\delta_{m-1} + \overline{a_2}\delta_m && (10.104)\\
0 &= R^* Q_1^{m+1} - R Q_1^{m+1,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1} && (10.105)
\end{aligned}
$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1} \tag{10.106}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$
\begin{aligned}
0 &= (c_2\overline{x_2} - \overline{a_2 x_2}) Q_1^m(x_2) - (c_2 x_2 - \overline{a_2 x_2}) Q_1^m(\overline{x_2}) && (10.107)\\
& - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m && (10.108)
\end{aligned}
$$

We start by extracting the $[x_2^1]$ coefficient

$$
\begin{aligned}
& c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} - c_2 Q_{1,2}^{m,0} + c_2 Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 && (10.109)\\
& c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 && (10.110)
\end{aligned}
$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geq 2$

$$c_2 Q_{1,2}^{m,n+2} - \overline{a_2} Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0 \tag{10.111}$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\widehat{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0 \tag{10.112}$$

Substituting this back into the above equation gives

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n} \tag{10.113}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 38. *For integers $m, n \geq 0$ we have the following relations*

$$b_1 Q_{1,2}^{m+2,n} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n} \tag{10.114}$$
$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n} \tag{10.115}$$

We can sum up those equations to obtain functional equations

Corollary 39. *For integers $m \geq 0$ we have the following relations*

$$Q_1^m - \widehat{Q}_1^m = b_1 Q_1^{m+2} \tag{10.116}$$
$$\widehat{Q}_1^m - \widehat{\widehat{Q}}_1^m = b_1 \widehat{Q}_1^{m+2} \tag{10.117}$$
$$x_1^2(Q_2^n - \widehat{Q}_2^n) = b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n}) \tag{10.118}$$
$$x_1^2(\widehat{Q}_2^n - \widehat{\widehat{Q}}_2^n) = b_1(\widehat{Q}_2^n - \widehat{Q}_{1,2}^{1,n} x_1 - \widehat{Q}_{1,2}^{0,n}) \tag{10.119}$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q_1^1 - Q_1^0) \tag{10.120}$$
$$x_1^2(\widehat{Q} - \widehat{\widehat{Q}}) = b_1(\widehat{Q} - x_1 \widehat{Q}_1^1 - \widehat{Q}_1^0) \tag{10.121}$$

Corollary 40. *For integers $n \geq 0$ we have the following relations*

$$Q_2^n - \widehat{Q}_2^n = b_2 Q_2^{n+2} \tag{10.122}$$

$$\widehat{Q}_2^n - \widehat{\widehat{Q}}_2^n = b_2 \widehat{Q}_2^{n+2} \tag{10.123}$$

$$x_2^2(Q_1^m - \widehat{Q}_1^m) = b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \tag{10.124}$$

$$x_2^2(\widehat{Q}_1^m - \widehat{\widehat{Q}}_1^m) = b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \tag{10.125}$$

$$x_2^2(Q - \widehat{Q}) = b_2(Q - x_2 Q_2^1 - Q_2^0) \tag{10.126}$$

$$x_2^2(\widehat{Q} - \widehat{\widehat{Q}}) = b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0) \tag{10.127}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0,0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0,0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$Q_{1,2}^{0,0} = 1 + t a_1 a_2 Q_{1,2}^{1,1} \tag{10.128}$$

$$Q_{1,2}^{1,1} = t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2}) \tag{10.129}$$

We also have

$$b_1 b_2 Q_{1,2}^{2,2} = b_1(Q_{1,2}^{2,0} - \widehat{Q}_{1,2}^{2,0}) \tag{10.130}$$

$$= b_1 Q_{1,2}^{2,0} - b_1 \widehat{Q}_{1,2}^{2,0} \tag{10.131}$$

$$= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} + \widehat{\widehat{Q}}_{1,2}^{0,0} \tag{10.132}$$

together with

$$b_1 b_2 Q_{1,2}^{2,0} = b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \tag{10.133}$$

$$b_1 b_2 Q_{1,2}^{0,2} = b_1 Q_{1,2}^{0,0} - b_1 \widehat{Q}_{1,2}^{0,0} \tag{10.134}$$

We we have

$$b_1b_2Q^{1,1}_{1,2} = t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \tag{10.135}$$

$$= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} \tag{10.136}$$

$$- b_1\widehat{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \tag{10.137}$$

$$= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} \tag{10.138}$$

$$- (b_1+1)\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \tag{10.139}$$

$$= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \tag{10.140}$$

We can finally write

$$b_1b_2Q^{0,0}_{1,2} = b_1b_2 + ta_1a_2b_1b_2Q^{1,1}_{1,2} \tag{10.141}$$

$$= b_1b_2 + t^2a_1a_2(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \tag{10.142}$$

Let $G(b_1,b_2)$ stand for $Q(0,0,b_1,b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2}) \tag{10.143}$$

or written as a linear combination

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2 \tag{10.144}$$

We can divide out by a_1a_2 and obtain the equation

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2 \tag{10.145}$$

from which we can read off

$$f(a_1,a_2,t) = c_1c_2 - a_1a_2t^2 \tag{10.146}$$

and

$$G = \frac{(c_1c_2 - t^2(a_2G_1 + a_1G_2 - G_{1,2})}{f(a_1,a_2,t)} \tag{10.147}$$

together with $B(fG) = c_1c_2$ Also, we can solve for Q_i^j and friends, in terms of various $[b_k^l]$ extractions, so overall it should lead to a relationship of the form $B(FQ) = 0$.

10.3 The almost-full model: missing steps E and W

10.4 The full model: all steps available

10.5 Conjectured relationships depending on $S_{1,2}^{-1,-1}$

As we have seen whenever $S_{1,2}^{-1,-1} = 0$, α and G tend to go together in the equations, so in the end we have a solution that depends on $\beta(G)$ multiplied by G_1, G_2 and/or $G_{1,2}$. In the end this leads to a rational function solution for G in terms of G_1, G_2 and $G_{1,2}$. It's then easy to conjecture that this happens whenever we can have such an algebraic relationship. The other part of the conjecture is that we'd have a linear relationship between G, G_1, G_2 and $G_{1,2}$ when $S_{1,2}^{-1,-1} \neq 0$.

Chapter 11

Epilogue

11.1 Discussion

In this work we examined such and such things. This led to so and so stuff. Finally we see problems in this and that area. Some things remain a mystery.

11.2 Future Work

Obviously we need to solve the remaining problems. We also need to solve/explain the mysteries presented so far. Additionally we need to rigourize the subject some more and bring in some more abstraction to make the methods more powerful. Full treatment of flarvoured problems.

11.3 Conclusion

This work solved the challenges it presented itself and uncovered more challenges that await a good tackling.

Chapter 12

Miscellaneous Things

If we could prove the validity of one of the functional equations in the corollaries (5th row in each corollary) we would then be able to extract coefficients and obtain the formulas in the first corollary. However, since we have a direct proof already, this section is kind of irrelevant now. We'll just keep it for now because it has some interesting math. The steps from now on are almost exactly the same as those that you'd have to do to prove the Dyck path formula. We'll prove the last one involving a b_1 coefficient. First we have

$$K = 1 - tS = 1 - t(x_1 + \overline{x_1})(x_2 + \overline{x_2}) \tag{12.1}$$

$$K_1 = c_1 - t\overline{x_1}S_1^{-1} = c_1 + \widehat{K}_1 \tag{12.2}$$

$$K_2 = c_2 - t\overline{x_2}S_2^{-1} \tag{12.3}$$

$$K_{1,2} = c_1c_2 - t\overline{x_1x_2} = c_1c_2 + \widehat{K}_{1,2} \tag{12.4}$$

We also need the coefficient extractions

$$[x_1^0]: \quad Q_1^0 - c_2Q_{1,2}^{0,0} - ta_1S_1^{-1}Q_1^1 = \overline{a_2} \tag{12.5}$$

$$[x_1^0]: Q_1^1 - tS_1^{-1}Q_1^2 - tS_1^{-1}Q_1^0 + t\overline{x_2}Q_{1,2}^{2,0} + t\overline{x_2}Q_{1,2}^{0,0} = 0 \tag{12.6}$$

$$-t\overline{x_1}S_1^{-1}Q_1^2 + t\overline{x_1x_2}Q_{1,2}^{2,0} = t\overline{x_1}S_1^{-1}Q_1^0 - t\overline{x_1x_2}Q_{1,2}^{0,0} - \overline{x_1}Q_1^1 \tag{12.7}$$

which we'll use soon. We first start with

$$KQ = K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \tag{12.8}$$

$$= (c_1 + \widehat{K}_1) Q_1^0 + K_2 Q_2^0 - (c_1 c_2 + \widehat{K}_{1,2}) Q_{1,2}^{0,0} + \overline{a_1 a_2} \tag{12.9}$$

$$K\widehat{Q} = \widehat{K}_1 \widehat{Q}_1^0 + K_2 \widehat{Q}_2^0 - \widehat{K}_{1,2} \widehat{Q}_{1,2}^{0,0} + \overline{a_2} \tag{12.10}$$

Then we have

$$KQ - K\widehat{Q} = \widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2 (Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + (c_1 Q_1^0 - c_1 c_2 Q_{1,2}^{0,0} + \overline{a_1 a_2} - \overline{a_2}) \tag{12.11}$$

$$= \widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2 (Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + t b_1 S_1^{-1} Q_1^1 \tag{12.12}$$

Also

$$\widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) = \widehat{K}_1 b_1 Q_1^2 - \widehat{K}_{1,2} b_1 Q_{1,2}^{2,0} \tag{12.13}$$

$$= b_1 (\widehat{K}_1 Q_1^2 - \widehat{K}_{1,2} Q_{1,2}^{0,0} \tag{12.14}$$

$$= b_1 (-t\overline{x_1} S_1^{-1} Q_1^2 + t\overline{x_1 x_2} Q_{1,2}^{2,0}) \tag{12.15}$$

$$= b_1 (t\overline{x_1} S_1^{-1} Q_1^0 - t\overline{x_1 x_2} Q_{1,2}^{0,0} - \overline{x_1} Q_1^1) \tag{12.16}$$

Next

$$K_2 (Q_2^0 - \widehat{Q}_2^0) = (c_2 - t\overline{x_1}^2 S_2^{-1}) \overline{x_2} b_1 (Q_2^0 - Q_{1,2}^{0,0}) \tag{12.17}$$

$$= -t\overline{x_1}^2 \overline{x_2} b_1 S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0}) + c_2 b_1 \overline{x_1}^2 (Q_2^0 - Q_{1,2}^{0,0}) \tag{12.18}$$

We can finally combine these expressions into

$$x_1^2 (KQ - K\widehat{Q}) = b_1 x_1^2 (t S_1^{-1} Q_1^1 + t\overline{x_1} S_1^{-1} Q_1^0 - t\overline{x_1 x_2} Q_{1,2}^{0,0} - \overline{x_1} Q_1^1 - t\overline{x_1}^2 \overline{x_2} S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0}) + c_2 \overline{x_1}^2 (Q_2^0 - Q_{1,2}^{0,0})) \tag{12.19}$$

We cancel out the b_1 and we have the equation

$$KQ - K(x_1 Q_1^1 + Q_1^0)) = t x_1 S_1^{-1} (x_1 Q_1^1 + Q_1^0) - (t x_1 \overline{x_2} Q_{1,2}^{0,0} + x_1 Q_1^1 + t\overline{x_2} S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0})) - c_2 (Q_2 - Q_{1,2}^{0,0}) \tag{12.20}$$

Note that

$$K + tx_1 S_1^{-1} = 1 - t\overline{x_1} S_1^{-1} \tag{12.21}$$

Once we expand the KQ term on the LHS we get

$$c_1 Q_1^0 - t\overline{x_1} S_1^{-1} Q_1^0 + c_2 Q_2^0 - t\overline{x_2} S_1^{-1} Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} + t\overline{x_1 x_2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \tag{12.22}$$

$$= (1 - t\overline{x_1} S_1^{-1})(x_1 Q_1^1 + Q_1^0) - x_1 Q_1^1 + t\overline{x_1 x_2} Q_{1,2}^{0,0} - t\overline{x_2} S_2^{-1} Q_2^0 + c_2 (Q_2^0 - Q_{1,2}^{0,0}) \tag{12.23}$$

and after careful book-keeping all the terms cancel out one by one. This completes the long-winded proof of the identity.

Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.